

Connections

➡ Building on

- Nucleophilic substitution at saturated carbon **ch15**
- Conjugate addition **ch22**
- Controlling stereochemistry **ch14**, **ch32**, & **ch33**
- Oxidation and reduction **ch23**
- Chemistry of Si and Sn **ch27**
- Aromatic heterocycles **ch29** & **ch30**
- Cycloadditions **ch34**
- Rearrangements **ch35** & **ch36**
- Radicals and carbenes **ch37** & **ch38**

Arriving at

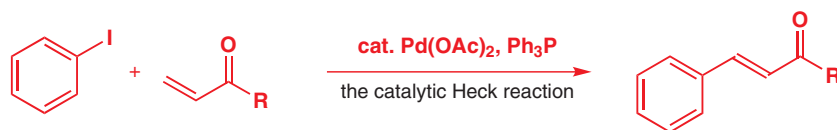
- Transition metals form organic compounds
- The structure of σ and π complexes and the meaning of η numbers
- The bonding is described with the usual orbitals
- Most stable complexes have 18 valence electrons
- Metals catalyse 'impossible' reactions
- Oxidative insertion, reductive elimination, and ligand migration from metal to carbon are key steps
- Carbon monoxide inserts into metal–carbon bonds
- Palladium is the most important metal
- C–C, C–O, and C–N bonds can be made with Pd catalysis
- Cross-coupling of two ligands is common
- Allyl cation complexes are useful electrophiles

➡ Looking forward to

- Asymmetric synthesis **ch41**
- The chemistry of life, especially nucleic acids **ch42**

Transition metals extend the range of organic reactions

Some of the most exciting reactions in organic chemistry make use of transition metals, and in recent years three Nobel prizes have been awarded for work in this area. How about this example? It is a **Heck reaction**, which allows nucleophilic addition to an unactivated alkene. Just a catalytic amount of palladium is needed to make the reaction go: the most useful organometallic reactions are those in which the metal acts catalytically.



Reagents and complexes containing transition metals are important in modern organic synthesis because they allow apparently impossible reactions to occur easily. Their chemistry complements traditional functional group chemistry and significantly broadens the range of

■ True to our beliefs that explanations are more important than facts, we have given mechanisms for most reactions. You should understand that it is much more difficult to be certain about these mechanisms than in some other areas of chemistry: a great deal of work goes into establishing mechanisms for important organometallic transformations, but many remain speculative. The mechanisms we give are meant to help you understand what is going on and may well turn out not to be correct in every detail.

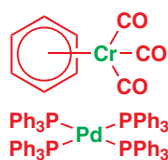
reactions a chemist can expect to use to make molecules. This chapter introduces the concepts of metal–ligand interaction, describes the most important reactions that can occur while ligands are bound to the metal, and demonstrates the power of organometallic chemistry in synthesis. The efficiency of transition metal-catalysed reactions means that they are routinely used in industrial synthesis. It is important that you understand the rules by which organometallic chemistry works.

The 18 electron rule

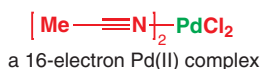
There is a contradiction in what is required of a metal complex for it to be useful to us. Initially, it will need to be stable and have a long enough lifetime to enable study and, ideally, storage. But once it enters the reaction vessel, stability is a disadvantage: instead we want reactivity. Our ideal catalyst is a complex that is stable in the resting state, but quickly becomes activated in solution—perhaps by loss of a ligand—so that it can interact with the substrate. Fortunately, there is a simple guide to the stability of transition metal complexes: **the 18 electron rule**. If a complex satisfies the 18-electron rule it means that the metal at the centre of the complex has the noble gas configuration of 18 electrons in the valence shell, and the complex is likely to be stable. The requirement for 18 electrons comes from the need to fill one ‘s’ orbital, five ‘d’ orbitals, and three ‘p’ orbitals with two electrons in each. The 18 electrons we need can come from those the metal already possesses plus those donated by any coordinating ligands.

The table below gives you the number of valence electrons each metal starts with before it acquires any ligands. Notice that the ‘new’ group numbers 1–18 give you the answer without any calculation. The most important are highlighted.

Group	IVB (4)	VB (5)	VIB (6)	VIIB (7)	VIII (8, 9, and 10)			1A (11)
Number of valence electrons	4	5	6	7	8	9	10	11
3d electrons	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
4d electrons	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag
5d electrons	Hf	Ta	W	Re	Os	Ir	Pt	Au



18-electron complexes



a 16-electron Pd(II) complex

Metals to the left-hand side of this list obviously need many more electrons to make up the magic 18. Chromium, for example, forms stable complexes with a benzene ring, giving it six electrons, and three molecules of carbon monoxide, giving it two each: $6 + 6 + 2 + 2 + 2 = 18$. Palladium is happy with just four triphenylphosphines (Ph_3P) giving it two each: $10 + 2 + 2 + 2 + 2 = 18$.

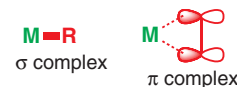
You may already know from your study of inorganic chemistry that there are exceptions to the 18-electron rule, particularly among complexes of Ti, Zr, Ni, Pd, and Pt, which can all form stable 16-electron complexes. The important 16-electron Pd(II) complex with two chlorides and two acetonitriles (MeCN) will feature heavily in this chapter. The so-called platinum metals Ni, Pd, and Pt are extremely important in catalytic processes, as you will see later on. Their stable 16-electron configuration results from a high-energy vacant orbital caused by the complex adopting a square planar geometry.

Ligands can be attached in many different ways



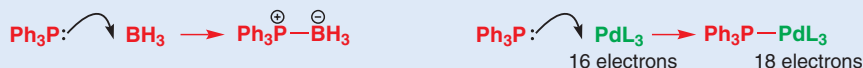
Transition metals can have a number of ligands attached to them and each ligand can be attached in more than one place. This affects the reactivity of the ligand and the metal because each additional point of attachment means the donation of more electrons. We can show the number of atoms involved in bonding to the metal by a **hapto number** η . A simple Grignard reagent is η^1 (pronounced ‘eta-one’) as the magnesium is attached only to one carbon atom. A metal–alkene complex is η^2 because both carbon atoms of the alkene are equally involved in bonding to the metal. In these cases the η designation is not very informative as there are no alternatives, and it is usually omitted.

The bonding in these two complexes is very different. In the first there is a simple σ bond between the metal and the alkyl group as in a Grignard reagent $R-MgBr$ and this type of complex is called a σ complex. In the alkene complex, bonding is to the p orbitals only. There are no σ bonds to the metal, which sits in the middle of the π bond in between the two p orbitals. This type of complex is called a π complex.



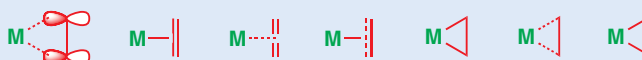
Representing bonds in transition metal complexes

It is difficult to know exactly how to draw the bonding in metal complexes and there are often several different acceptable representations. There is no problem when the metal forms a σ bond to atoms such as Cl or C as the simple line we normally use for covalent bonds means exactly what it says. The problems arise with ligands that form σ bonds by donating both their electrons, and with π complexes. Everyone writes phosphine–borane complexes with two charges but we normally draw the same sort of bond between a phosphine and, say, Pd as a simple line with no charges.

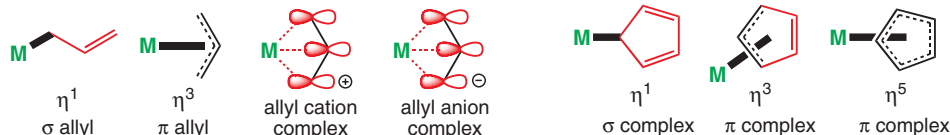


You will sometimes see π complexes drawn with simpler dotted lines going to the middle of the π bond, sometimes with dotted π bonds, and sometimes with bonds (simple or dotted) going to the ends of the old π bond. These are all acceptable as the bonding is complex, as you will see. We might almost say that the ambiguity is helpful: we often don't know either the exact nature of the bonding or the number of other ligands in the complex. In the diagrams in this section we have shown the main bond from metal to ligand as a heavy line in the simplest representation but we also offer alternatives with simple and dotted bonds. Don't worry about this—things will become clearer as the chapter develops. When you have to draw the structure of a complex but you don't know the exact bonding, just draw a line from metal to ligand.

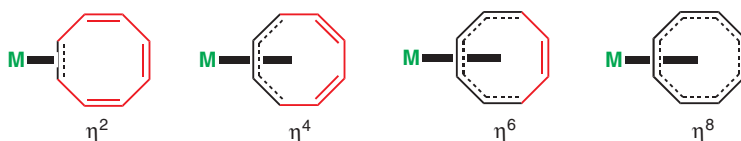
different acceptable ways
to draw π complexes



These labels are useful where there is a choice of type of bonding, as with allylic ligands. The metal can either form a σ bond to a single carbon (hence η^1), or form a π complex with the p orbitals of all three carbons of the allyl system—this would be η^3 . If the π complex is made from an allyl cation, the ligand has two electrons, but it has four if it is made from an allyl anion. Similarly, a cyclopentadienyl anion can act as a σ ligand (η^1), an allyl ligand (η^3), or, most usually, as a cyclopentadienyl ligand (η^5). The distinction is very important for electron counting as these three different situations contribute two, four, or six electrons, respectively, to the complex.



Neutral ligands can also bond in a variety of ways. Cyclooctatetraene can act as an alkene (η^2), a diene (η^4), a triene (η^6), or a tetraene (η^8), and the reactivity of the ligand changes accordingly. These are all π complexes with the metal above or below the black portion of the ring and with the thick bond to the metal at right angles to the alkene plane.

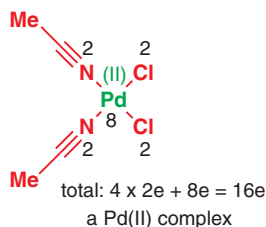
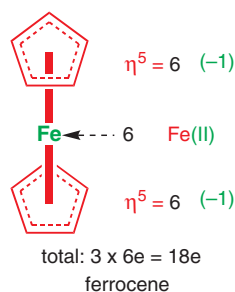
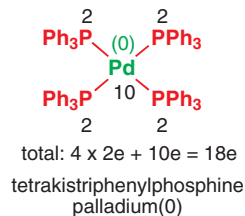


To determine the number of electrons around the transition metal in a complex the valence electrons from the metal ion are added to those contributed by all the ligands. The numbers of electrons donated by various classes of ligands are summarized in the table. Anions such as halides, cyanide, alkoxide, hydride, and alkyl donate two electrons, as do neutral ligands with a lone pair such as phosphines, amines, ethers, sulfides, carbon monoxide, nitriles, and

isonitriles. Unsaturated ligands can contribute as many as eight electrons and can be neutral or negatively charged. If the overall total is 18, then the complex is likely to be stable. If the overall total is less than 18 the complex is called *coordinatively unsaturated*.

Ligand characteristics	Formal charge	Electrons donated
anionic ligands		
Cl^- Br^- I^- CN^- OR^- H^- alkyl^-	-1	2
neutral σ -donor ligands		
	0	2

Organic ligands	Hapto number	Formal charge	Electrons donated
unsaturated σ - or π -donor ligands			
aryl, σ -allyl	η^1	-1	2
alkenes	η^2	0	2
π -allyl cation	η^3	+1	2
π -allyl anion	η^3	-1	4
diene—conjugated	η^4	0	4
dienyls, cyclopentadienyls (anions)	η^5	-1	6
arenes, trienes	η^6	0	6
trienyls, cycloheptatrienyls (anions)	η^7	-1	8
cyclooctatetraene	η^8	0	8
carbene, nitrene, oxo	η^1	0	2



Electron counting helps to explain the stability of metal complexes

Counting electrons in most complexes is simple using the table of ligand characteristics above in conjunction with the table on p. 1070. Take tetrakis(triphenylphosphine) palladium(0): each neutral phosphine donates two electrons, making a total of eight, and palladium still has its full complement of 10 valence electrons as it is in the zero oxidation state. Overall, the complex has a total of 18 electrons and is a stable complex—in fact too stable: it has to lose a PPh_3 ligand before it can enter into reactions.

All of the different classes of ligands listed in the table above can be treated in this way. In ferrocene, the cyclopentadienyl ligands contribute six electrons each and have a formal negative charge, shown in green, which means that the iron in ferrocene is in the +2 oxidation state and will have six valence electrons left. The total for the complex is again 18 and ferrocene is an extremely stable complex.

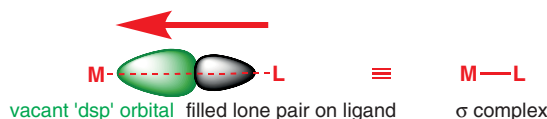
The useful complex $(\text{MeCN})_2\text{PdCl}_2$ has palladium in the +2 oxidation state because of its two chlorine atoms and the number of electrons is eight for the Pd(II) oxidation state and another two each from the four ligands, making 16 in all. This complex does not fulfil the 18-electron rule and is stable yet reactive.

The oxidation state of metals in complexes

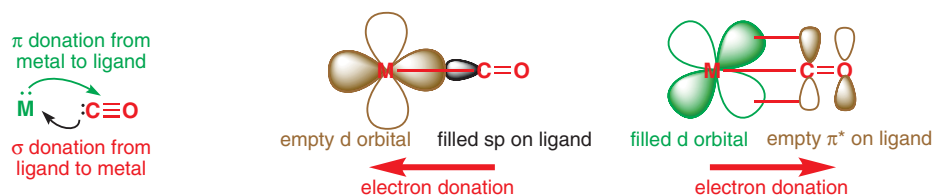
As well as the problem of bond drawing, there is a potential problem over oxidation states too. You can either say that ferrocene is a complex of Fe(II), having two fewer electrons than the normal eight, with two cyclopentadienyl anions contributing six electrons each, or you can say that it is a complex of Fe(0), having eight electrons, with two cyclopentadienyl ligands each contributing five electrons. The simplest approach is to say that a metal is in the (0) oxidation state unless it has σ bonds to ligands such as Cl, AcO , or Me that form bonds with shared electrons. Neutral ligands such as Ph_3P that provide two of their own electrons do not affect the oxidation state of the metal.

Bonding and reactions in transition metal complexes

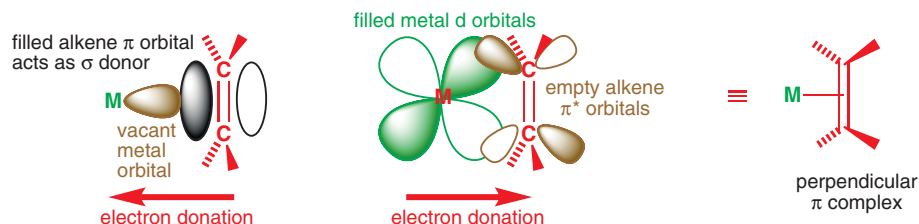
The majority of ligands have a lone pair of electrons in a filled sp^n type orbital that can overlap with a vacant metal 'dsp' orbital, derived from the vacant d, p, and s orbitals of the metal, to form a conventional two-electron two-centre σ bond. Ligands of this type increase the electron density on the central metal atom.



A bonding interaction is also possible between any filled d orbitals on the metal and vacant ligand orbitals of appropriate symmetry such as π^* orbitals. This leads to a reduction of electron density on the metal and is known as **back-bonding**. An example would be a complex with carbon monoxide. Many metals form these complexes and they are known as **metal carbonyls**. The ligand (CO) donates the lone pair on carbon into an empty orbital on the metal while the metal donates electrons into the low-energy π^* orbital of CO. Direct evidence for this back-bonding is an increase in the C—O bond length and a lowering of the infrared stretching frequency from the population of the π^* orbital of the carbonyl.



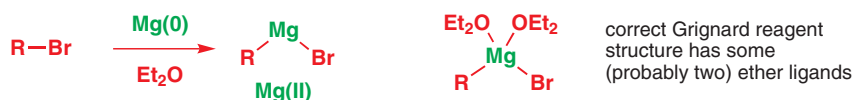
When an unsaturated ligand such as an alkene approaches the metal sideways to form a π complex, similar interactions lead to bonding. The filled π orbitals of the ligand bond to empty d orbitals of the metal, while filled d orbitals on the metal bond to the empty π^* orbitals of the ligand. The result is a π complex with the metal–alkene bond perpendicular to the plane of the alkene. The bond has both σ and π character.



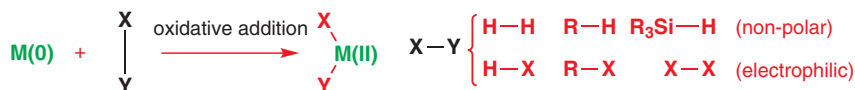
Coordination to a metal by any of these bonding methods changes the reactivity of the ligands dramatically and this is exploited in the organometallic chemistry we will be discussing in the rest of the chapter. You do not need to understand all the bonding properties of metal complexes but you need to be able to count electrons, to recognize both σ and π complexes, and to realize that complexes show a balance between electron donation and electron withdrawal by the metal.

Oxidative addition inserts metal atoms into single bonds

Potential ligands that do not have a lone pair or filled π -type orbital are still able to interact with transition metal complexes but only by breaking a σ bond. This is the first step in a wide variety of processes and is described as **oxidative addition** because the formal oxidation state of the transition metal is raised by two, for example M(0) to M(II), in the process. This is the result of having two extra ligands bearing a formal negative charge. You have seen this process in the formation of Grignard reagents (Chapter 9).



The number of coordinated ligands also increases by two so the starting complex is usually in low oxidation state (0 or 1; the diagram shows 0) and **coordinationally unsaturated**, that is, it has an empty site for a ligand and, say, only 16 electrons, like $(\text{MeCN})_2\text{PdCl}_2$, whereas the product is usually **coordinationally saturated**, that is, it cannot accept another ligand unless it loses one first.



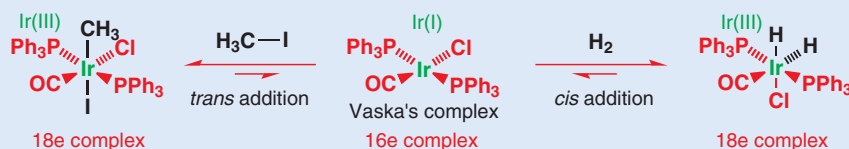
➡ You will see why Wilkinson's catalyst and its derivatives are important catalysts of homogeneous hydrogenation in Chapter 41.

Oxidative addition occurs for a number of useful neutral species, including molecular hydrogen, carbon–hydrogen bonds, and silanes as well as polarized bonds or electrophilic species containing at least one electronegative atom. The resulting species with metal–ligand bonds allow useful chemical transformations to occur. Important examples include the oxidative addition of Pd(0) to aryl iodides and the activation of Wilkinson's catalyst for hydrogenation in solution by oxidative addition to a hydrogen molecule.



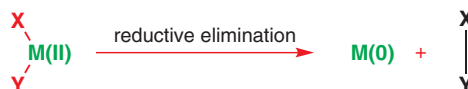
Vaska's complex

There are a number of possible mechanisms for oxidative addition and the precise one followed depends on the nature of the reacting partners. Vaska's complex $[\text{Ir(PPh}_3)_2\text{COCl}]$ has been extensively studied and it reacts differently with hydrogen and methyl iodide. Hydrogen is added in a *cis* fashion, consistent with concerted formation of the two new iridium–hydrogen bonds. The 16e (count them!), d^8 , Ir(I) complex becomes a new 18e, d^6 , Ir(III) species. With methyl iodide the kinetic product is that of *trans* addition, which is geometrically impossible from a concerted process. Instead, an S_N2 -like mechanism is followed involving nucleophilic displacement of iodide followed by ionic recombination.

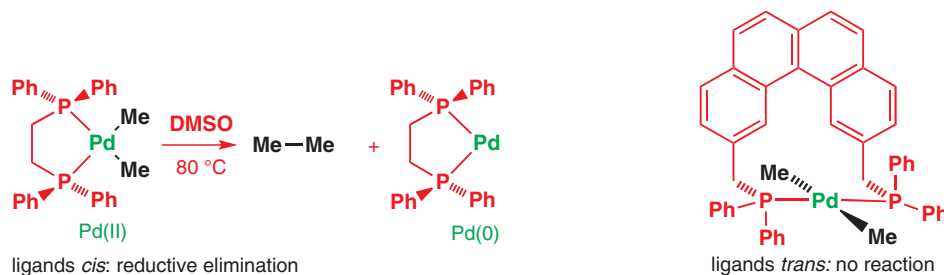


Reductive elimination removes metal atoms and forms new single bonds

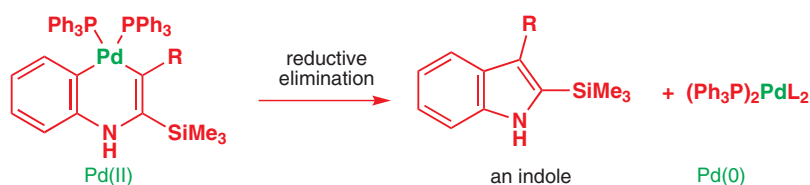
If we want to use organometallic chemistry to make organic compounds other than those containing metals, we must be able to remove the ligands from the coordination sphere of the metal at the end of the reaction. Neutral organic species such as alkenes, phosphines, and carbon monoxide can simply dissociate in the presence of other suitable ligands but those that are bound to the metal with shared electrons require a more active process. Fortunately, most reactions that occur around a transition metal are reversible and so the reverse of oxidative addition, known as **reductive elimination**, provides a simple route for the release of neutral organic products from a complex. Our general reaction shows M(II) going to M(0) , releasing X-Y . These two ligands were separate in the complex but are bound together in the product. A new X-Y σ bond has been formed.



The ligands to be eliminated must be *cis* to one another for reductive elimination to occur. This is because the process is concerted. Two examples from palladium chemistry make this point clear. Warming in DMSO causes ethane production from the first palladium complex because the two methyl groups are *cis* in the square planar complex. The more elaborate second bisphosphine forces the two methyl groups to be *trans* and reductive elimination does not occur under the same conditions.



Of course, no one wants to make ethane that way (if at all) but many other pairs of ligands can be coupled by reductive elimination. Reductive elimination is one of the most important methods for the removal of a transition metal from a reaction sequence, leaving a neutral organic product. We will see many examples as the chapter develops but here is an indole synthesis that depends on a reductive elimination at palladium as a last step. In the starting material, palladium has two σ bonds sharing electrons with C, and is Pd(II). In the reaction the two C substituents bond together to form the indole ring and a Pd(0) species is eliminated.

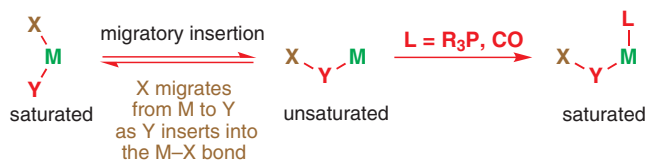


L and L_n

We know that the Pd product here must carry more than just the two ligands it is left with after the reductive elimination, but the scheme we have here doesn't tell us what they are. It's quite conventional in such cases—where we are more interested in the structure of the organic product than the remains of the complex—to indicate a general metal ligand 'L', or 'L_n' for an undefined number of unknown ligands.

Migratory insertion builds ligand structure

Two ligands can also react together to produce a new complex that still has the composite ligand attached to the metal, ready for further modification. This process involves migration of one of the ligands from the metal to the other ligand, and insertion of one of the ligands into the other metal–ligand bond. It is known as **migratory insertion**. The insertion process is reversible and, as the metal effectively loses a ligand in the process, the overall insertion may be driven by the addition of extra external ligands (L) to produce a coordinatively saturated complex. As with reductive elimination, a *cis* arrangement of the ligands is required and the migrating group (X) retains its stereochemistry (if it has any) during the migration.



Wilkinson's catalyst is used in homogeneous hydrogenation of alkenes. The catalyst is soluble in many organic solvents such as EtOH, chloroform, or some hydrocarbons. The alkene complexes with the metal and migratory insertion forms an alkyl metal complex by hydrogen transfer. The next step, reductive elimination, usually follows rapidly to give the alkane and a complex that adds a hydrogen molecule to regenerate the catalyst.

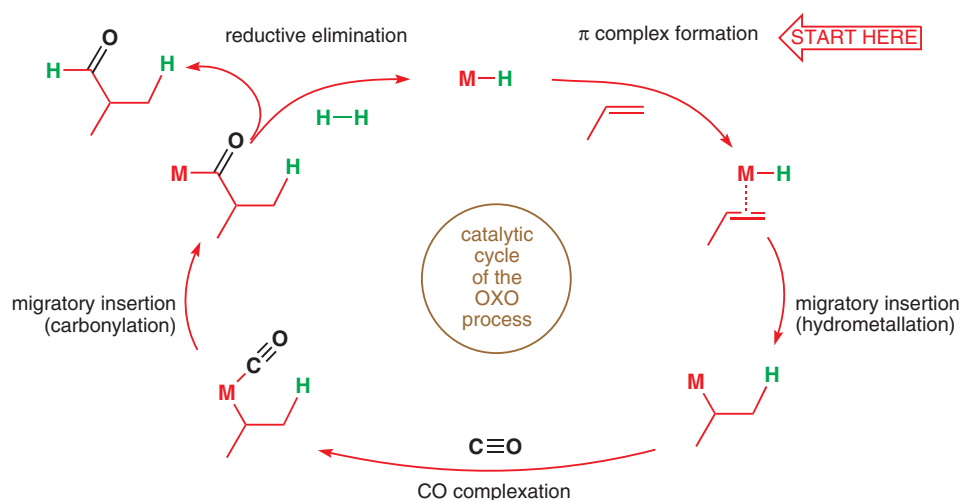
➡ Migration normally occurs with retention; see Chapter 36.

Carbon monoxide incorporation extends the carbon chain

Carbonylation (the addition of carbon monoxide to organic molecules) is an important industrial process as carbon monoxide is a convenient one-carbon feedstock and the resulting metal-acyl complexes can be converted into aldehydes, acids, and their derivatives. The **OXO process** is the hydroformylation of alkenes such as propene and uses two migratory insertions to make higher value aldehydes. Although a mixture is formed, this is acceptable from very cheap and abundant starting materials. Here the metal complex is a catalyst, not a stoichiometric reagent.

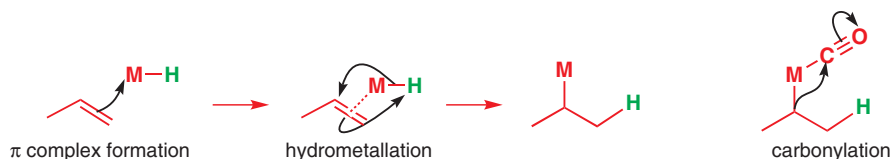


A catalytic cycle (going clockwise from the top) shows the various stages of alkene coordination, hydrometallation (migratory insertion) to produce an alkyl metal species, coordination of carbon monoxide followed by another migratory insertion, and finally reductive cleavage with hydrogen to produce the metal-hydride intermediate, which is then ready for another cycle. The steps leading to the other regioisomeric aldehyde and the ligands on the metal are omitted for clarity.



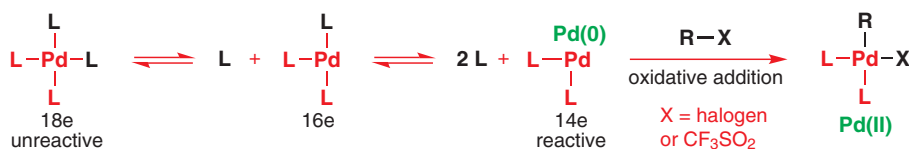
Interactive mechanism for the OXO process

The mechanisms of the two key migratory insertion steps are worth discussion. **Hydrometallation** occurs by initial π complex formation followed by addition of the metal to one end of the alkene and hydrogen to the other. Both of the possible regioisomers are formed. The **carbonyl insertion reaction** is another migration from the metal to the carbon atom of a CO ligand.

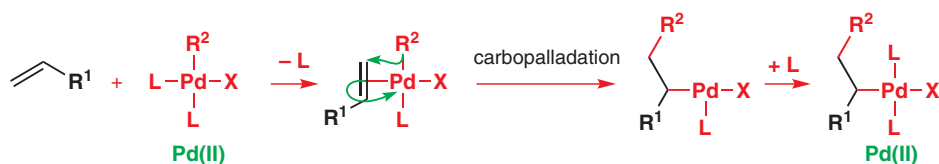


Insertion reactions are reversible

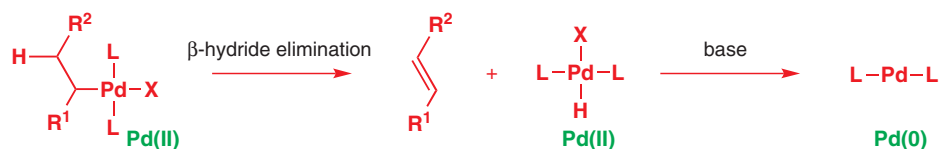
The reverse process, **decarbonylation**, is also fast but can be arrested by maintaining a pressure of carbon monoxide above the reaction mixture. The reverse of hydrometallation involves the elimination of a hydride from the adjacent carbon of a metal alkyl to form an alkene complex.



The resulting Pd–R σ bond in such complexes is very reactive, especially towards carbon–carbon π bonds. Thus an alkene in the reacting system will lead to coordination followed by migratory insertion into the palladium–carbon σ bond. Like hydrometallation this process is a migratory insertion and is called **carbopalladation** because carbon and palladium become attached to the ends of the alkene system. There is no change in oxidation state during this process, although the ligands (often phosphines) must dissociate to allow coordination of the alkene and associate to provide a stable final 16-electron product.



With some metals the process of olefin coordination and insertion may continue, leading to polymerization, but with palladium the metal is expelled from the molecule by a β -hydride elimination reaction and the product is an alkene, plus a Pd(II) complex. For the whole process to be catalytic, this Pd(II) product of β -hydride elimination must be converted to a Pd(0). This occurs in the presence of base, which removes HX from the palladium(II) species. This is another example of reductive elimination: one that forms a hydrogen halide rather than a carbon–carbon or carbon–hydrogen bond, as you saw earlier.



The speed of the β -hydride elimination (which is intramolecular and very fast) means that the original substrate for the oxidative addition reaction must be chosen with care—the presence of hydrogen at an sp^3 carbon in the β position must be avoided. Thus, substrates for oxidative addition reactions in palladium chemistry are frequently vinylic, allylic, or aromatic and never ethyl or *n*-propyl.

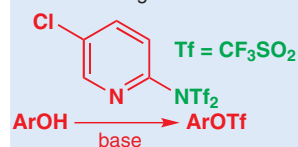
The Heck reaction couples together an organic halide or triflate and an alkene

All the individual steps outlined above combine to make up the catalytic pathway in the **Heck reaction** with which we started the chapter. The Heck reaction couples an alkene with an organic halide or triflate $\text{R}^1\text{-X}$ to form a new alkene. The R^1 group in $\text{R}^1\text{-X}$ can be aryl, vinyl, or any alkyl group without β hydrogens on an sp^3 carbon atom. The group X can be a halogen (Br or I) or triflate (OSO_2CF_3). The alkene can be mono- or disubstituted and can be electron-rich, -poor, or -neutral. The base need not be at all strong and can be Et_3N , NaOAc , or aqueous Na_2CO_3 . The reaction is very accommodating!

Triflates

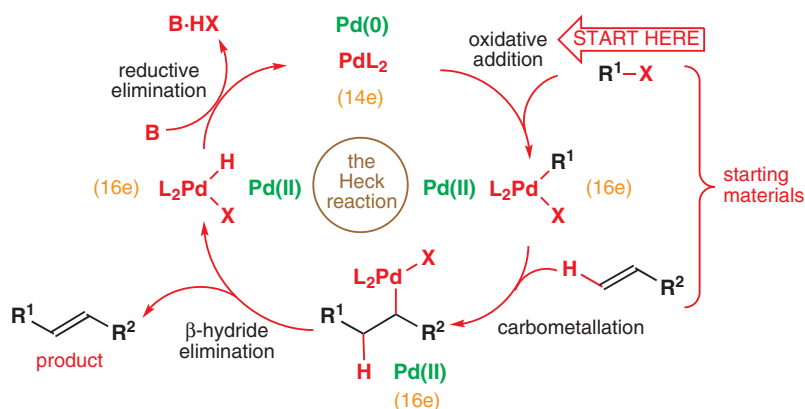
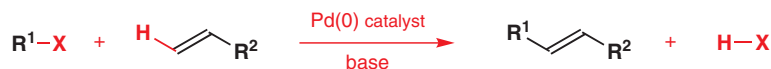
The triflate (trifluoromethanesulfonate) anion, CF_3SO_3^- , or TfO^- , is an excellent, non-basic leaving group. It is often used as an oxygen-based alternative to halides, and metals will insert into the C– OSO_2CF_3 bond. Triflates, particularly aryl and vinyl triflates, can be made conveniently with Comins' reagent.

Comins' reagent



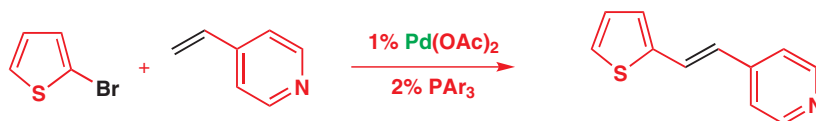
Interactive mechanism for the Heck catalytic cycle

the Heck reaction



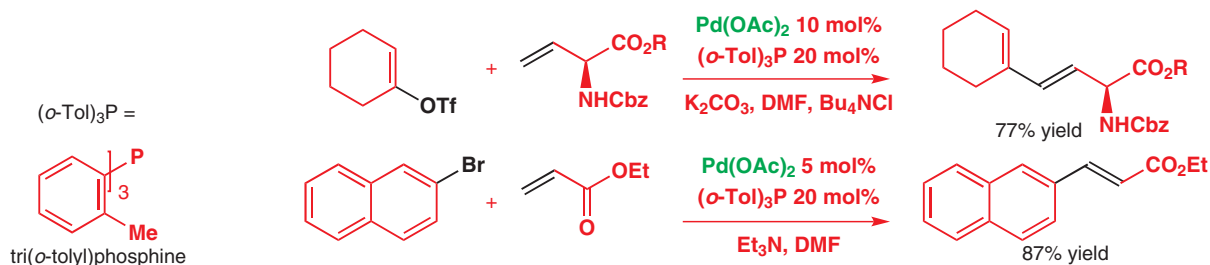
This palladium-catalysed addition of aryl, vinyl, or substituted vinyl groups to organic halides or triflates is one of the most synthetically useful palladium-catalysed reactions. The method is very efficient, and carries out a transformation that is difficult by more traditional techniques. The mechanism involves the oxidative addition of the halide, insertion of the olefin, and elimination of the product by a β -hydride elimination process. A base then regenerates the palladium(0) catalyst. The whole process is a catalytic cycle.

Here is the Heck reaction at work coupling two heterocyclic substrates. Easy chemistry to do, but impossible without a Pd catalyst.



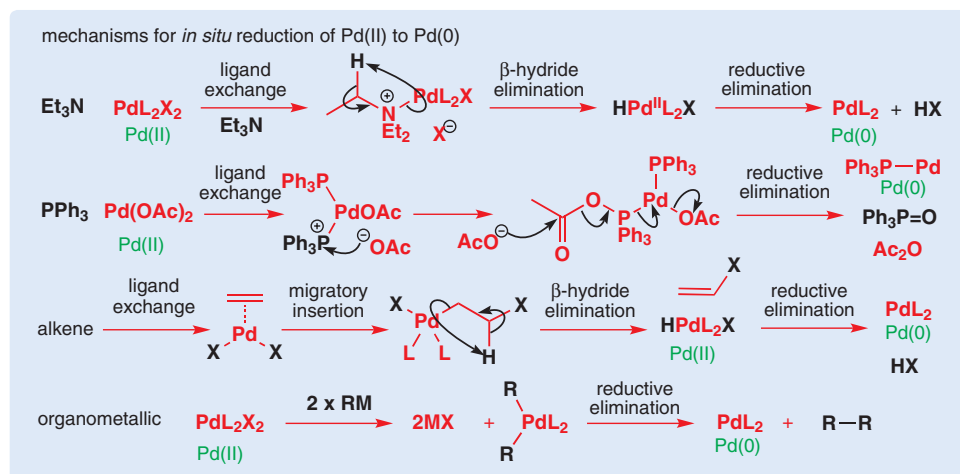
Notice the regioselectivity: unlike the carbonylation on p. 1077, the Heck reaction favours one isomer, and when the alkene is polarized by an electron-withdrawing group the new C–C bond forms at the other end of the alkene. Notice also in this example and those below that the Pd is added as Pd(II), not Pd(0): the box below explains how this works.

The mild conditions of the Heck reaction mean that protected amino acids can be made without any racemization. The two examples below use a more hindered analogue of triphenylphosphine, but the mechanism is the same.

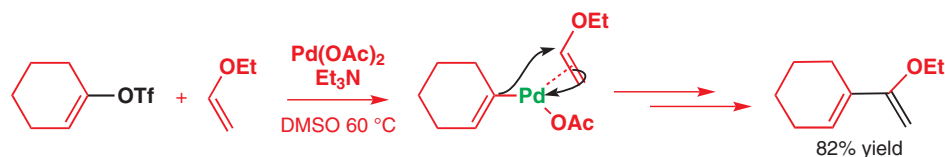


In situ formation of Pd(0) by reduction of Pd(II)

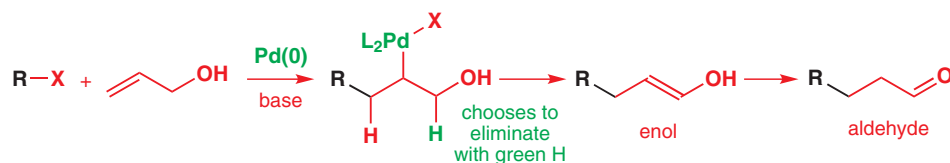
In reactions requiring palladium(0), formation of the active complex may be achieved more conveniently by reduction of a palladium(II) complex, for example $\text{Pd}(\text{OAc})_2$. Any phosphine may then be used in the reaction, without the need to synthesize and isolate the corresponding palladium(0)–phosphine complex. The reduction of palladium(II) to palladium(0) can be achieved with amines, phosphines, alkenes, and organometallics such as DIBAL-H, butyllithium, or trialkylaluminium. The mechanisms are worth surveying as they illustrate the basic steps of organometallic chemistry.



In contrast, electron-donating groups such as ethers lead to attack at the end of the alkene substituted by oxygen to produce the 1,1-disubstituted product. These reactions must be dominated by the interaction of the filled π orbital of the alkene with an empty d orbital on Pd. In the example below, the Heck reaction works even in the absence of a phosphine ligand.

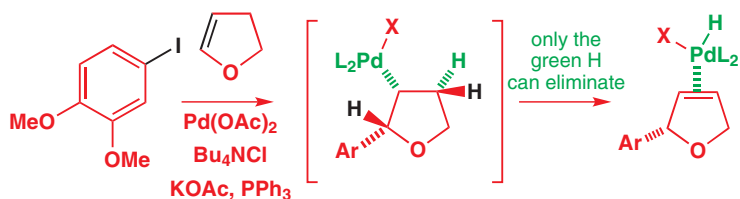


Because β -hydride elimination is reversible, when there is a choice the more stable of the possible alkenes usually results. The reaction of allylic alcohols is particularly important as the more stable of the two alkenes is the enol and a carbonyl compound is formed.

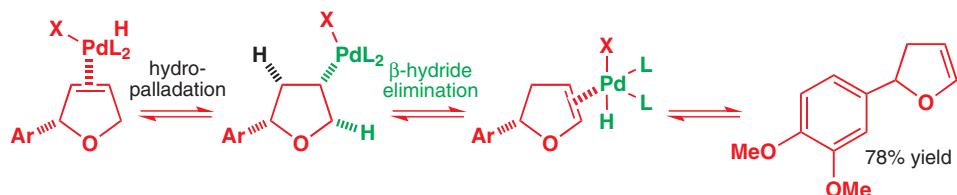


Hydropalladation–dehydropalladation can lead to alkene isomerization

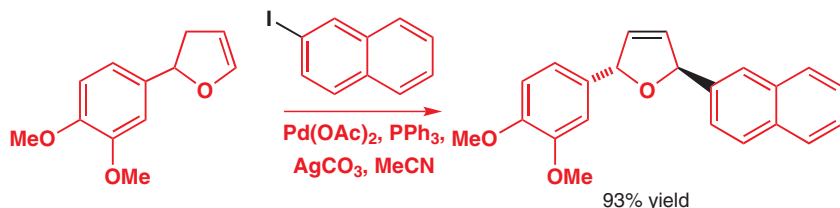
Reversible β -hydride eliminations provide a mechanism for interconverting regioisomers of an alkene, and the following reaction sequence also illustrates another point about the reductive elimination step: it is a *syn* elimination, and the C–Pd and C–H bonds have to eclipse one another for the Pd–H bond to form. Oxidative addition of the aryl iodide to a palladium(0) complex, formed from Pd(OAc)₂ by reduction, gives the active palladium(II) complex ArPdOAcL₂. Carbopalladation occurs as expected on an electron-rich alkene to give the product of aryl addition to the oxygen end of the alkene in a *syn* fashion. β -Hydride elimination must occur away from the aryl group because there is only one C–H bond *syn* to the C–Pd bond. The alkene has moved one position round the ring.



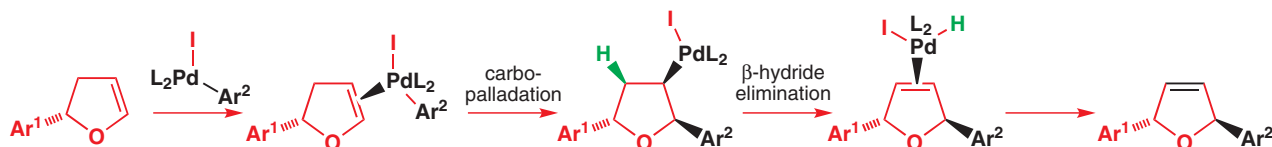
Hydropalladation in the reverse sense gives a new σ complex, which could eliminate either the black or the green hydrogens. Elimination of the green H gives the enol ether, which is the most stable alkene possible due to conjugation.



This product now undergoes a second Heck reaction involving naphthyl iodide:

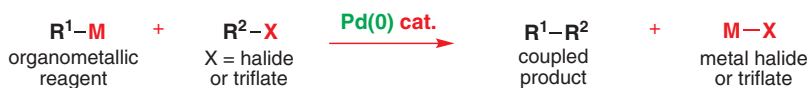


The initial mechanism is much the same. However, the enol ether has two diastereotopic faces: *syn* or *anti* to the aromatic substituent (Ar^1) introduced in the first step. Palladium is very sensitive to steric effects and generally forms less hindered complexes where possible, so the palladium(II) complexes the face of the enol ether *anti* to Ar^1 . This in turn controls all the subsequent steps, which must be *syn*, leading to a final product with *anti* stereochemistry. The requirement for *syn* β -hydride elimination also explains the regiochemical preference of the elimination. In the σ -bonded cyclic structure there is only one hydrogen (green) that is *syn* to the palladium; the one on the carbon bearing the naphthyl substituent is *anti* and cannot be eliminated. Further migrations of the alkene by hydropalladation are prevented by the silver carbonate, which rapidly removes iodide from the intermediate, preventing readdition of Pd-H to the alkene.

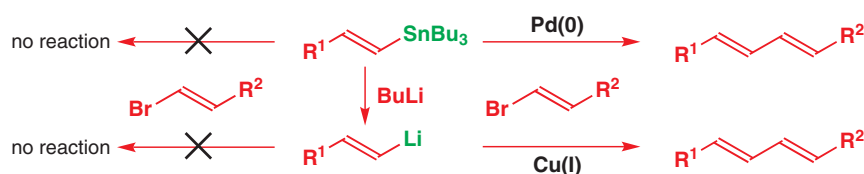


Cross-coupling of organometallics and halides

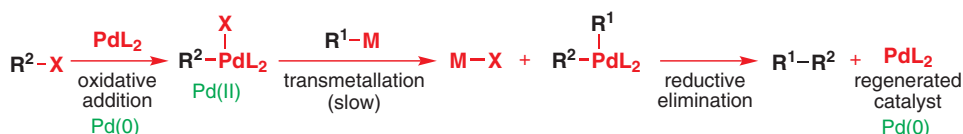
Other than β -hydride elimination, another important pathway by which palladium(II) intermediates can lead to neutral organic fragments is reductive elimination. This forms the basis of the mechanism for **cross-coupling reactions** between an organometallic reagent and an organic halide or triflate.



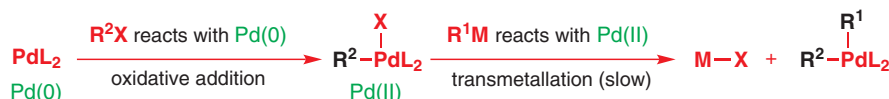
This is a reaction that seems very attractive for synthesis but, in the absence of a transition metal catalyst, the yields are very low. We showed in Chapter 27 how vinyl silanes can be made with control over stereochemistry and converted into lithium derivatives with retention. Neither of these vinyl metals couple with vinyl halides alone. But in the presence of a transition metal— Cu(I) for Li and Pd(0) for Sn —coupling occurs stereospecifically and in good yield.



The mechanism of palladium-catalysed cross-coupling starts, as in the Heck reaction, with oxidative addition of the halide or triflate to the initial palladium(0) phosphine complex to form a palladium(II) species. But the next step is new: it is a **transmetallation**, so-called because the nucleophile (R^1) is transferred from the metal in the organometallic reagent to the palladium and the counterion (X =halide or triflate) moves in the opposite direction. The new palladium(II) complex with two organic ligands undergoes reductive elimination to give the coupled product and the palladium(0) catalyst, ready for another cycle.



The reaction is important because it allows the coupling of two different components (R^1 and R^2), distinguished by being bonded either to the metal M or to the halide or triflate X . Both components form σ complexes with Pd but the halide partner (R^2X) bonds first by oxidative addition and this R^2-Pd bond must survive while the metal partner (R^1M) transfers R^1 to Pd by transmetalation. Once the two components are joined to the palladium atom, only the cross-coupled product can be formed. R^2X combines with $Pd(0)$ and R^1M with $Pd(II)$. There can then be no confusion. In contrast to the Heck reaction, here the metal defines the location of the new $C-Pd$ bond.

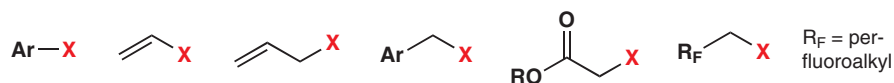


The halide partner (R^2X) must be chosen with care, as β -hydride elimination would decompose the first intermediate during the slow transmetalation step. The choice for R^2 is restricted to substituents without β -hydrogen atoms on sp^3 carbon atoms: vinyl, allyl, benzyl, and polyfluoroalkyl halides, triflates, and phosphates have all been coupled successfully. The organometallic reagent (R^1M) can be based on magnesium, zinc, copper, tin, silicon, zirconium, aluminium, or boron and the organic fragment can have a wide variety of structures as coupling is faster than β -hydride elimination.

R^1-M R^1 = almost anything including examples with β H
 $M = MgX, ZnX, Cu, SnR_3, SiR_3, ZrCp_2Cl, AlMe_2, B(OR)_2, BF_4^-$

Cp = cyclopentadienyl

R^2-X R^2 must not have β Hs that can eliminate $X = I, Br, (Cl), OTf, OPO(OR)_2$

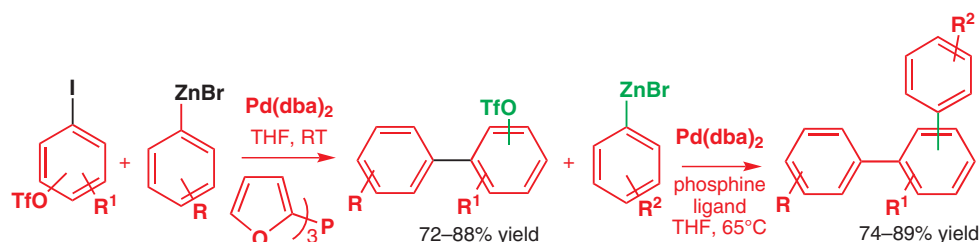


The difference in relative reactivity of aromatic iodides and triflates was exploited in this sequential synthesis of substituted terphenyls by repeated coupling with organozinc reagents. The more reactive iodide coupled at room temperature with palladium(0) and trifurylphosphine but warming to $65^\circ C$ was required for the triflate to participate in the second coupling.

■ There is a problem in naming the two partners. The halide partner (R^2X) is sometimes called the electrophile and the organometallic partner (R^1M) the nucleophile. These names describe the nature of the reagents rather than the mechanism of the reaction and we will not use them.

Named coupling reactions

Palladium-catalysed reactions involving organometallic partners based on B, Mg, Sn, and Zn are particularly important and are often referred to by the names of their discoverers: Suzuki coupling for B, Kumada coupling for Mg, Stille coupling for Sn, and Negishi coupling for Zn.



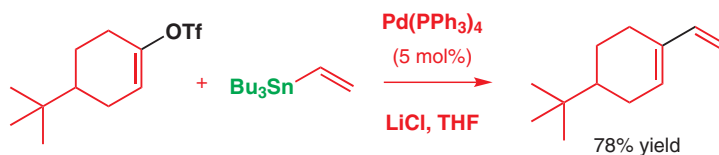
Organometallic Nobels

Organometallic chemistry has been a popular subject for the Nobel prize committee. In 1912 Grignard (Mg) won the award, in 1973 Wilkinson and Fischer for 'sandwich' compounds (such as ferrocene), in 2005 Chauvin, Grubbs, and Schrock for alkene metathesis, and in 2010 Heck, Negishi, and Suzuki (Stille had died in 1989) for transition-metal catalysed couplings.

In spite of the wide range of organometallic reagents that can be used there are two classes that have proved particularly popular because they are stable intermediates in their own right and can be prepared and purified separately before the coupling reaction. These cross-couplings are known by the names of the two chemists whose work made the reactions so valuable. The Stille coupling employs a stannane as the organometallic component (R^1M) while the Suzuki coupling relies on a boronic acid.

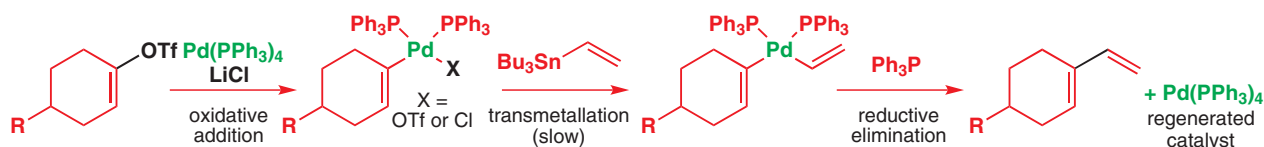
The Stille coupling uses stannanes as the organometallic component

Since its discovery in the late 1970s, the Stille coupling has been widely used for the coupling of both aromatic and vinylic systems.



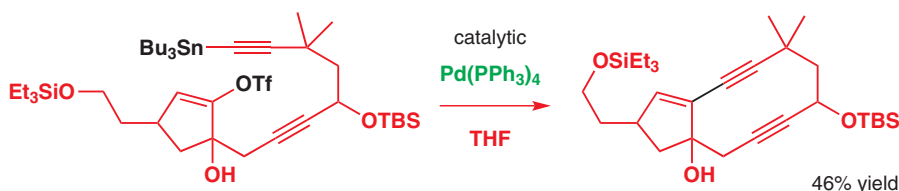
In the reactions of triflates, a source of halide (typically LiCl) is generally required since triflate is a counterion and may not bind to the metal as a ligand: chloride can take its place.

The mechanism involves the oxidative addition of the vinyl or aromatic triflate or halide to give an organopalladium intermediate. Transmetalation with the organostannane forms another organopalladium intermediate with two Pd–C σ bonds. A reductive elimination step releases the product and thereby regenerates the palladium(0) catalyst. Vinyl triflates can be made from enolizable aldehydes or ketones and aryl triflates from phenols, but the reaction also works with vinyl and aryl halides.

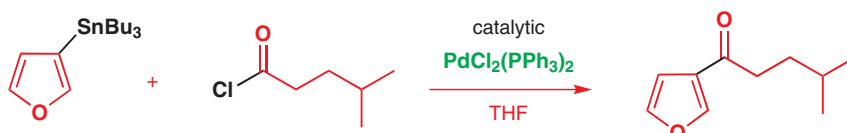


Interactive mechanism for the Stille coupling catalytic cycle

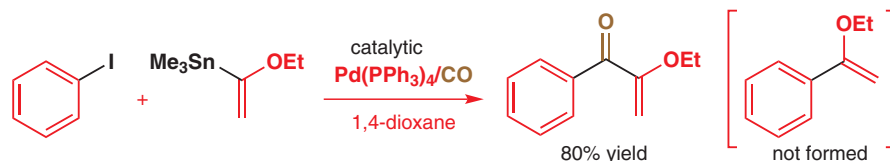
The Stille reaction is widely used to make bonds between sp^2 carbon atoms, but it also works with sp carbons: the example below is a challenging formation of a 10-membered ring containing two alkynes.



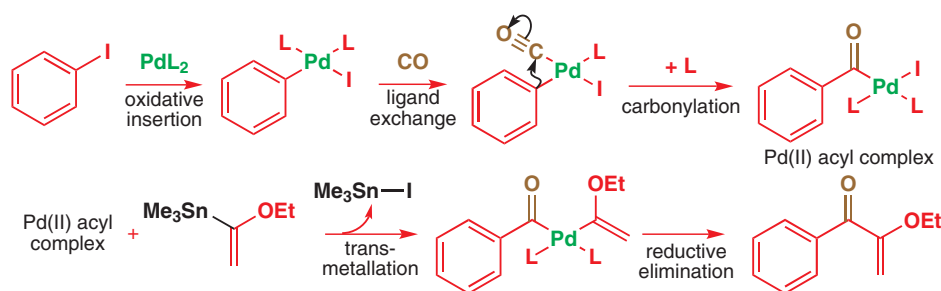
The Stille coupling may be combined with carbonylation in two ways. Acid chlorides may be used as substrates for the reaction with vinyl or aryl stannanes, although an atmosphere of carbon monoxide is frequently required to prevent decarbonylation after the oxidative addition step.



Simply performing a normal Stille reaction in the presence of carbon monoxide may also lead to carbonylated products. These reactions can take place in a CO saturated solution, under one atmosphere of pressure. Using these conditions, excellent yields of the carbonylated product can be obtained, without any of the normal coupling product being present.



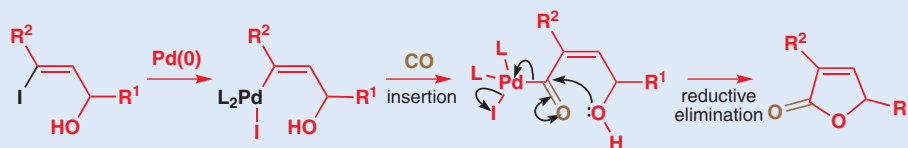
The mechanism follows that of a normal Stille coupling except that the carbon monoxide first exchanges for one of the phosphine ligands and then very rapidly inserts to produce an acyl palladium(II) complex. Transmetalation with the vinyl stannane in the usual way forms trimethylstannyl iodide and the key palladium complex carrying two carbon ligands. Transmetalation is always the slow step in these coupling reactions, allowing time for the carbon monoxide insertion. The final step—reductive elimination—releases the Pd(0) catalyst for the next cycle.



Interactive mechanism for the carbonylative Stille coupling catalytic cycle

Acyl palladium species react like activated acid derivatives

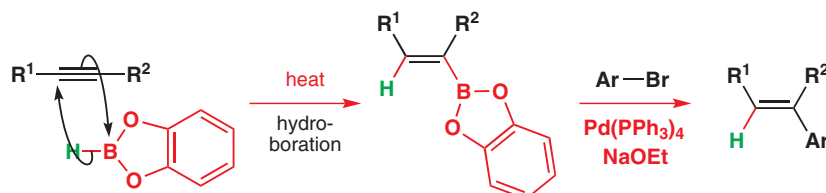
Carbonylation of a halide or triflate provides a direct route to a range of chain-extended acyl derivatives. A carbonyl group substituted with PdX (X = halide or triflate) is a reactive acylating agent, rather like an acid anhydride, as PdX is a good leaving group. Reaction with alcohols and amines gives esters and amides, while reduction with tributyltin hydride gives the aldehyde. Intramolecular attack by alcohols leads to lactones, as demonstrated in the conversion of a vinyl iodide into a 2H-furanone (butenolide). We will see more of these reactions later.



Interactive mechanism for palladium-catalysed carbonylative butenolide formation

The Suzuki coupling couples boronic acids to halides

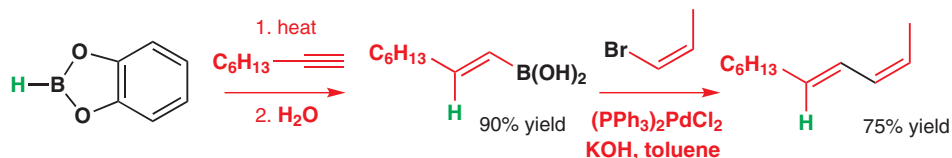
The Suzuki coupling of a boronic acid or ester with a vinyl or aryl halide or triflate is probably the most commonly used of all cross-coupling reactions. The original version, first reported in 1979, involved hydroboration of an alkyne with catecholborane, followed by palladium(0)-catalysed coupling of the resulting vinyl boronate with an aromatic iodide or bromide. The hydroboration is generally regioselective for the less hindered position and addition of boron and hydrogen occurs *cis* stereospecifically.



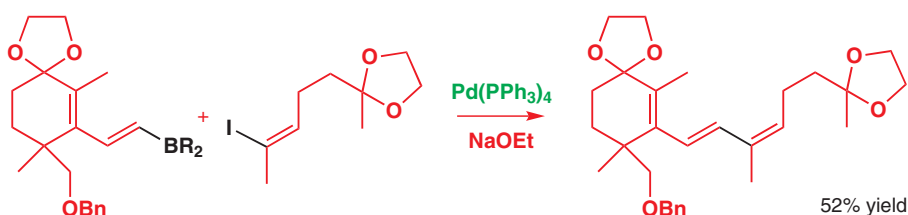
➡ Hydroboration is covered in Chapter 19, p. 446.

➡ Some methods for the synthesis of alkenes with control of double bond geometry were covered in Chapter 27.

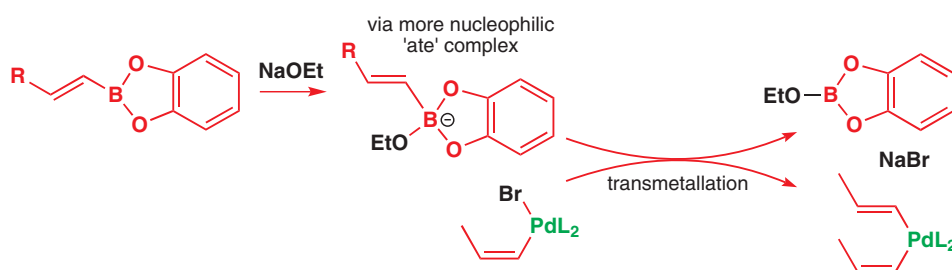
As in the Stille coupling, the geometry of both unsaturated components is preserved during the coupling so this is an excellent method for the stereoselective synthesis of dienes. Hydroboration of octyne followed by hydrolysis of the boronate gave exclusively the *E*-vinyl boronic acid. Coupling with the *Z*-vinyl bromide in toluene with palladium(0) catalysis with potassium hydroxide as the base gave the *E,Z*-diene in good yield. These dienes are very useful in the Diels–Alder reaction (Chapter 34).



This sort of reaction has been used in the synthesis of the unsaturated units of a range of natural products, including trisporol B. The key step is the stereocontrolled synthesis of an *E,Z*-diene. The geometry of both double bonds comes stereospecifically with retention of configuration from single geometrical isomers of the starting materials.

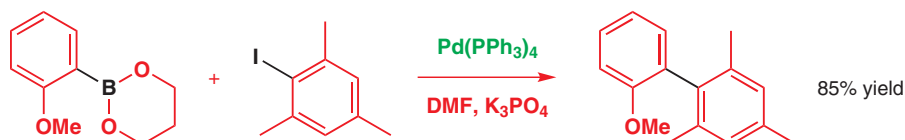


The mechanism of the Suzuki coupling is very similar to that of the Stille coupling. Oxidative addition of the vinylic or aromatic halide to the palladium(0) complex generates a palladium(II) intermediate. This then undergoes a transmetalation with the alkenyl boronate, from which the product is expelled by reductive elimination, regenerating the palladium(0) catalyst. The important difference is the transmetalation step, which explains the need for an additional base, usually sodium or potassium ethoxide or hydroxide, in the Suzuki coupling. The base accelerates the transmetalation step, leading to the borate directly, presumably via a more nucleophilic 'ate' complex.



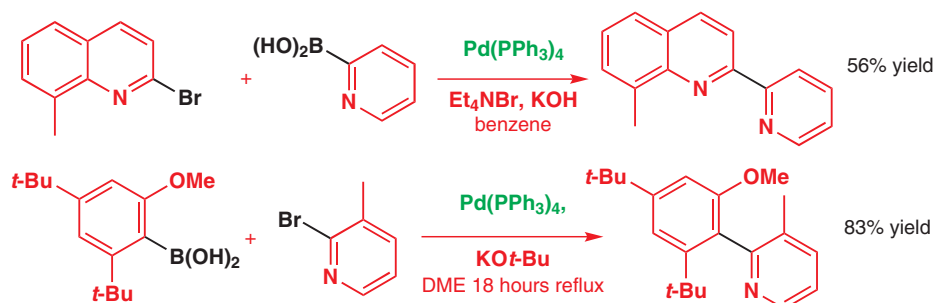
🖱 Interactive mechanism for the Suzuki coupling catalytic cycle

Sterically demanding substrates are tolerated well and Suzuki coupling is often used for aryl–aryl cross-couplings. This example has three *ortho* substituents around the newly formed bond (marked in black) and still goes in excellent yield.



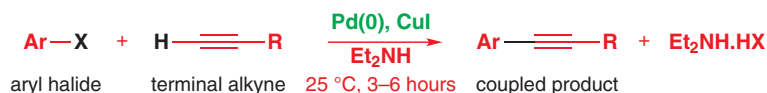
Aromatic heterocycles also couple well. The 2-position of a pyridine is very electrophilic and not at all nucleophilic (Chapter 29) but couplings at this position are fine with either the

halide or the boronic acid in that position. Clearly, it is a mistake to see either of these substituents as contributing a ‘nucleophilic carbon’. It is better to see the reaction as a coupling of two equal partners with the two substituents (the halide and the boronic acid) as control elements to ensure cross-coupling and prevent dimerization.

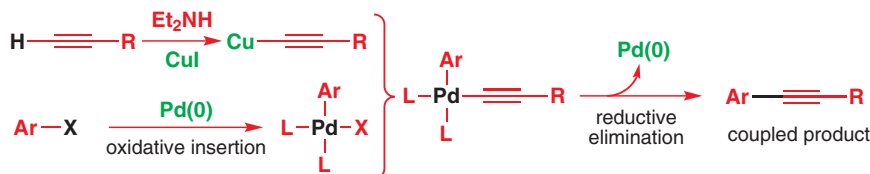


Coupling to alkynes: the Sonogashira reaction

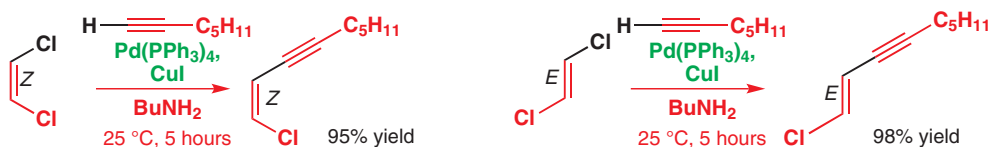
The coupling of terminal alkynes with aryl or vinyl halides under palladium catalysis is known as the **Sonogashira reaction** and is rather like the Heck reaction. It is a catalytic process, requiring a palladium(0) complex; it is performed in the presence of base, and generally uses copper iodide as a co-catalyst. One partner—the aryl or vinyl halide—is the same as in the Stille and Suzuki couplings but the alkyne needs no metal to activate it: the reaction works with the alkyne itself.



The mild conditions usually employed, frequently room temperature, mean that the reaction can be used with thermally sensitive substrates. By now, you should not be surprised by the mechanism! Oxidative addition of the organic halide gives a palladium(II) intermediate that undergoes transmetalation with the alkynyl copper (generated from the terminal alkyne, base, and copper iodide). Reductive elimination with coupling of the two organic ligands gives the product and regenerates the palladium(0) catalyst.

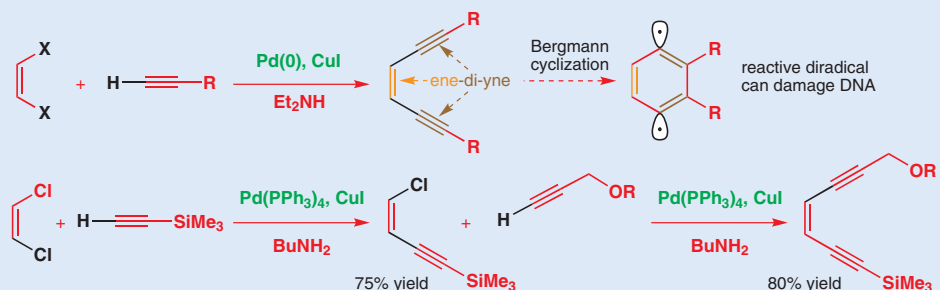


It is usually more convenient, as in the Heck reaction, to use a stable and soluble Pd(II) derivative such as bis(triphenylphosphine)palladium(II) chloride instead of Pd(0). This is rapidly reduced *in situ* to give a coordinatively unsaturated, catalytically active, palladium(0) species. The geometry of the alkene is generally preserved so that *cis* (*Z*) and *trans* (*E*) dichloroethylene give the two different geometrical isomers of the enyne below in >99% stereochemical purity as well as excellent yield.



Ene-diynes and the Bergmann cyclization

The Sonogashira reaction provides an important way to make the ene-diyne antibiotics. Symmetrical ene-diynes may be synthesized in one step from two molecules of a terminal alkyne and Z-dihaloethylene. The ene-diyne part of the molecule does the remarkable Bergmann cyclization to give a benzene diradical: the ene-diyne is able to penetrate DNA and the diradical is able to react with it, giving the compounds anticancer activity. To make the most biologically active compounds, however, the reaction is performed sequentially, allowing different functionality on each of the alkyne units.



• Palladium-catalysed coupling reactions: a summary

Coupling an aryl or vinyl halide with...	Typical example (X=I, Br, OTf)	See page	Name of reaction
an alkene		1079	Heck
an aryl or vinyl stannane		1084	Stille
an aryl or vinylboronic acid or ester		1085	Suzuki
an alkyne		1087	Sonogashira
an amine		1092 (later in Buchwald-this chapter)	Hartwig

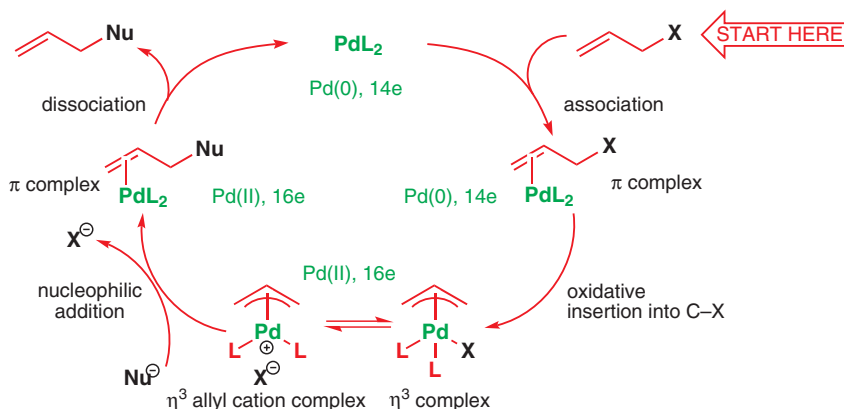
Allylic electrophiles are activated by palladium(0)

Allylic compounds with good leaving groups, such as bromide and iodide, are excellent allylating agents but they suffer from loss of regiochemistry due to competition between the direct S_N2 and S_N2' reactions. This problem was described in Chapter 24. In contrast, π -allyl

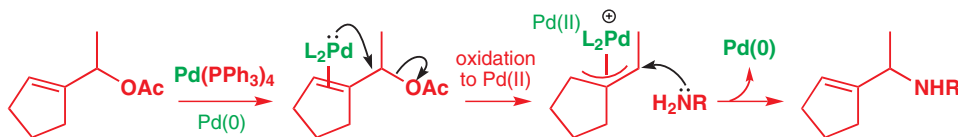
cation complexes of palladium allow both the stereochemistry and regiochemistry of nucleophilic displacement reactions to be controlled.



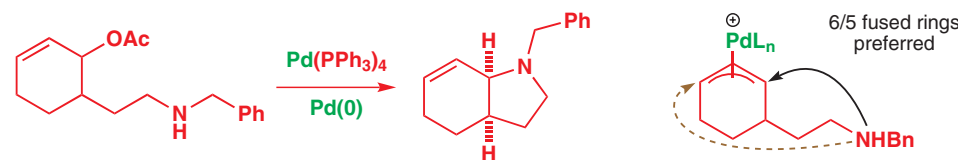
In addition, leaving groups (X) that are usually regarded as rather unreactive still work, which makes the purification and handling of the starting materials easier. Acetate ($X=\text{OAc}$) is the most commonly used leaving group, but a range of other functional groups ($X=\text{OCO}_2\text{R}$, $\text{OPO}(\text{OR})_2$, Cl , Br , OPh) will perform a similar role. The full catalytic cycle is shown below, with the intermediate π -allyl complex in equilibrium between the neutral version, which has the leaving group coordinated to palladium, and the cationic π -allyl complex.



Soft nucleophiles generally give the best results: stabilized enolates such as malonates, or cyanide, are best for carbon-carbon bond formation, but for C-X ($X=\text{O}$, N , S) bond formation the reaction is successful with alkoxides, amines, and thiolates (RS^-). In the example below an amine nucleophile attacks the allyl system to generate the more stable product with the double bond within the ring.



The intramolecular reaction works well to give heterocyclic rings—the regioselectivity is usually determined by the length of the chain and how far it can reach. Here a 6/5 fused product is preferred to a bridged product containing two seven-membered rings.

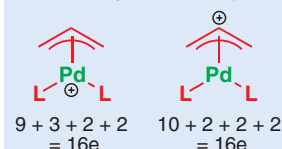


The reaction usually proceeds with *retention* of configuration at the reacting centre. As in $\text{S}_{\text{N}}2$ substitution reactions going with retention (Chapter 36), this actually suggests a double inversion. Coordination of Pd to the double bond of the allylic acetate occurs on the less hindered face opposite the leaving group and we can think of the oxidative addition step as an invertive nucleophilic displacement of the leaving group by a pair of Pd electrons. The nucleophile then adds to the face of the π -allyl Pd cation complex opposite the Pd. The net result is displacement of the leaving group by the nucleophile with retention. Thereafter, the

The Pd π -allyl cation complex

You can represent the palladium π -allyl cation complex in two ways. Either you draw a neutral allyl group complexed to Pd^+ or you draw an allyl cation complexed to neutral Pd. Although the counting is different (Pd^+ has only nine electrons: the neutral allyl has three, but the allyl cation only two), both come out as η^3 16-electron species, which is just as well as they are merely different ways of drawing the same thing.

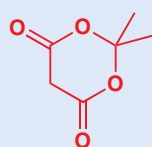
Pd π -allyl cation complex



Interactive mechanism for the π -allyl palladium-mediated coupling catalytic cycle

■ The arrows on the middle two diagrams are the best we can do to show how Pd(0) uses its electrons to get rid of the leaving group to become Pd(II), and how it accepts them back again when the nucleophile adds. They are not perfect: it is often difficult to draw precise arrows for organometallic mechanisms, but it is worth thinking about what is happening to the electrons in these steps, and curly arrows help us to do this.

Meldrum's acid has a very stable delocalized enolate: it is as acidic as a carboxylic acid (pK_a 4.97) and the unusual stability of the enolate comes from the fixed conformation of the two carbonyl groups.

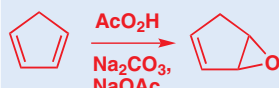


Meldrum's acid

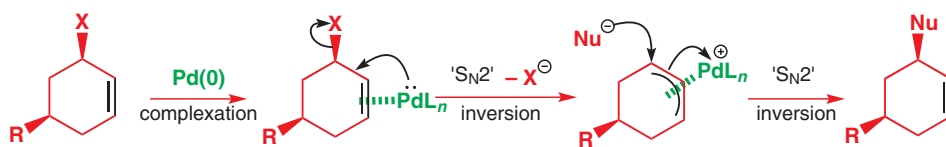
■ For many of the remaining schemes in the chapter we will ignore the additional ligands at palladium for simplicity's sake.

Making vinyl epoxides

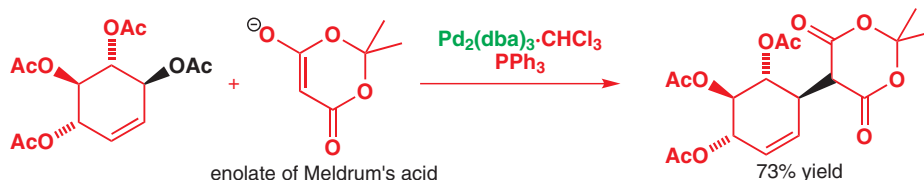
The synthesis of vinyl epoxides from dienes is mentioned in Chapter 19. The monoepoxide is formed first as the diene is more nucleophilic than the alkene in the monoepoxide. The main difficulty is that the monoepoxide rearranges with acid catalysis from the by-product, the carboxylic acid of the peroxyacid used in the epoxidation. The solution is simple: the mixture must be buffered to keep the acidity low.



nucleophile attacks from the less hindered face of the resulting π -allyl complex (that is, away from the metal), leading to overall retention of configuration.

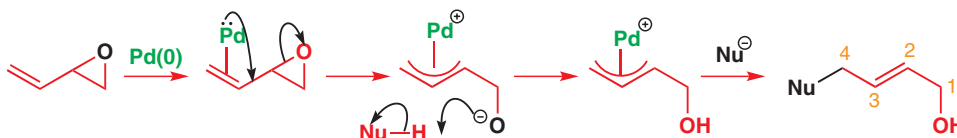


The reaction of this allylic acetate with the sodium salt of Meldrum's acid demonstrates the retention of configuration in the palladium(0)-catalysed process. The tetraacetate and the intermediate π -allyl complex are symmetrical, thus removing any ambiguity in the formation or reaction of the π -allyl complex and hence in the regiochemistry of the overall reaction.

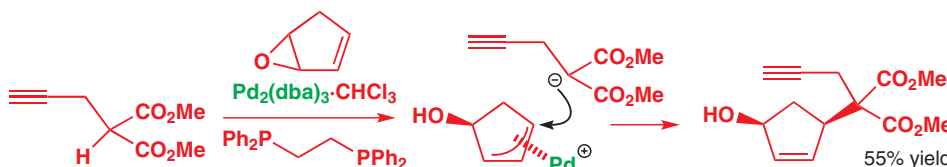


Vinyl epoxides provide their own alkoxide base

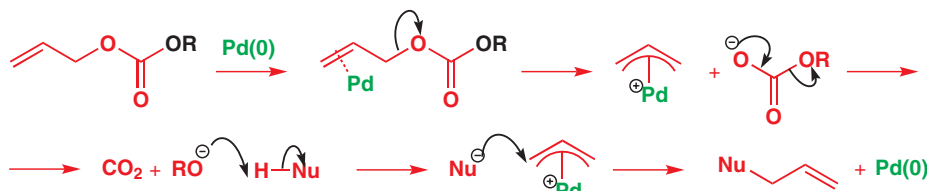
Vinyl epoxides and allylic carbonates are especially useful electrophiles because under the influence of palladium(0) they generate an alkoxide base, so no added base is required with these substrates. The overall reaction proceeds under almost neutral conditions—ideal with complex and sensitive substrates. The relief of strain in the three-membered ring drives the reaction with palladium(0) to produce the zwitterionic intermediate. Proton transfer activates the nucleophile, and attack at the less hindered end of the π -allyl palladium intermediate preferentially leads to overall 1,4-addition of NuH.



Retention of stereochemistry is demonstrated by the reaction of a substituted malonate with epoxycyclopentadiene. Palladium adds to the side opposite the epoxide so the nucleophile is forced to add from the same side as the OH group. This, no doubt, helps 1,4-regioselectivity.

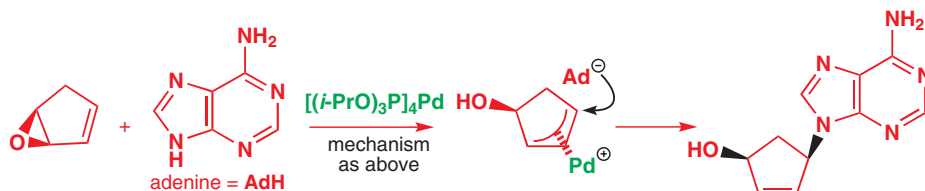
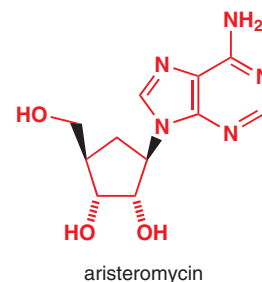


Allylic carbonates produce the required alkoxide by decarboxylation of the carbonate anion that is displaced in the formation of the π -allyl palladium intermediate. Deprotonation activates the nucleophile, which rapidly traps the π -allyl palladium complex to give the allylated product, regenerating the palladium(0) catalyst.

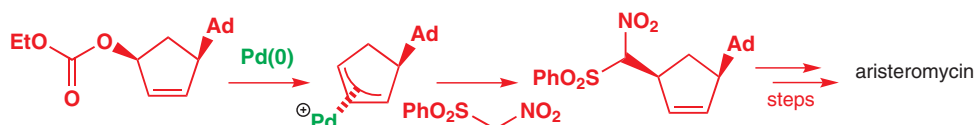


Trost and his group have used both of these palladium-catalysed alkylations in a synthesis of aristeromycin from epoxycyclopentadiene. The *cis* stereochemistry of this carbocyclic nucleotide analogue is of paramount importance and was completely controlled by retention of configuration in both substitutions.

The first reaction is between epoxycyclopentadiene and adenine, one of the heterocyclic building blocks of nucleic acids, and follows the mechanism we have just described to give a *cis*-1,4-disubstituted cyclopentene.

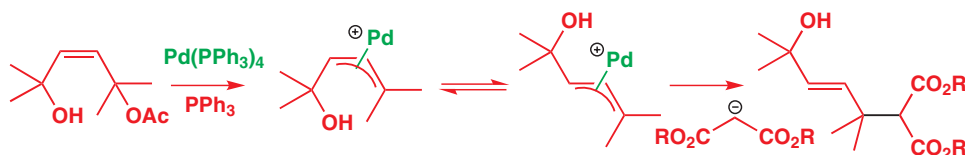


The alcohol is then activated by conversion into the carbonate, which reacts with phenylsulfonylnitromethane, and could later be converted into an alcohol. Once again, retention of stereochemistry during the palladium-catalysed substitution gives the *cis* product.

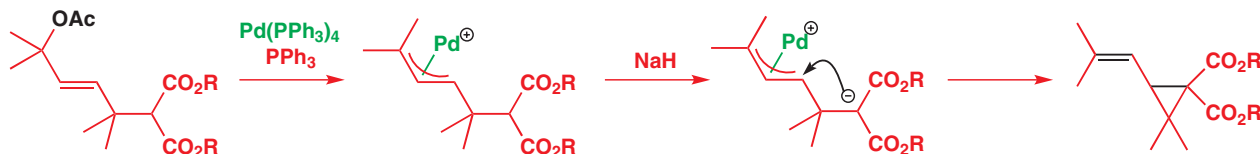


Intramolecular alkylations make rings

π -Allyl intermediates may also be used in cyclization reactions, including the synthesis of small and medium-sized rings using an intramolecular nucleophilic displacement. Three-membered rings form surprisingly easily, taking advantage of the fact that the leaving group can be remote from the nucleophile. The precursors can also be prepared by allylic alkylation. The sodium salts of malonate esters react with this monoacetate under palladium catalysis at the less hindered end to give the allylic alcohol.



Acetylation activates the second alcohol to displacement so that the combination of sodium hydride as base and palladium(0) catalyst leads to cyclization to the cyclopropane.

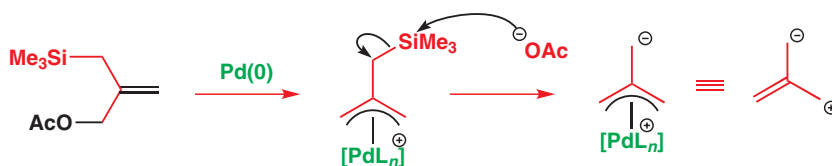


Palladium can catalyse cycloaddition reactions

The presence of five-membered rings such as cyclopentanes, cyclopentenones, and dihydrofurans in a wide range of target molecules has led to a variety of methods for their preparation. One of the most successful of these is the use of trimethylenemethane [3 + 2] cycloaddition, catalysed by palladium(0) complexes. The trimethylenemethane unit in these reactions is derived from 2-[(trimethylsilyl)methyl]-2-propen-1-yl acetate, which is at the same time an allyl silane and an allylic acetate. This makes it both a weak nucleophile and an electrophile in the presence of palladium(0). Formation of the palladium π -allyl complex is followed by removal of the trimethylsilyl group by nucleophilic attack of the

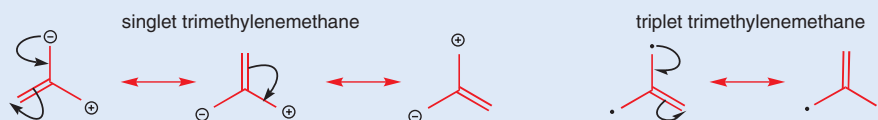
➡ Cycloadditions were described in Chapter 34.

resulting acetate ion, thus producing a zwitterionic palladium complex that can undergo cycloaddition reactions.

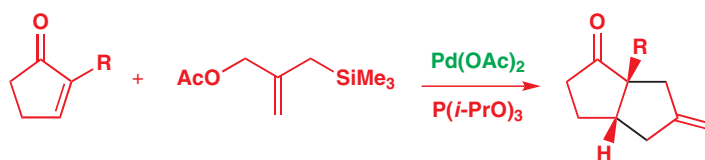


Trimethylenemethane

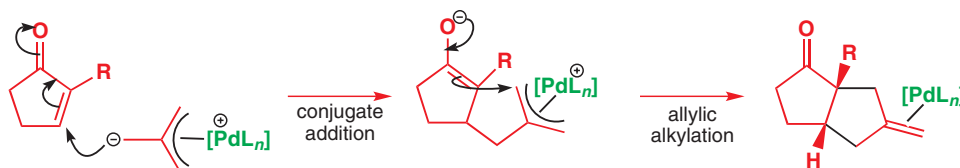
The symmetrical molecule with three CH₂ groups arranged trigonally about a carbon atom is interesting theoretically. It could have a singlet structure with two charges, both of which can be delocalized, but no neutral form can be drawn. Alternatively, it could be a triplet with the two unpaired electrons equally delocalized over the three CH₂ groups. This form is probably preferred and the singlet form is definitely known only as the palladium complex we are now describing. You might compare the singlet and triplet structures of trimethylenemethane with those of carbenes in Chapter 38.



The normal way to do the cycloadditions is to react the complex with an alkene bearing electron-withdrawing substituents that make the substrate prone to Michael-type conjugate addition. Cyclopentenones illustrate the reaction nicely.



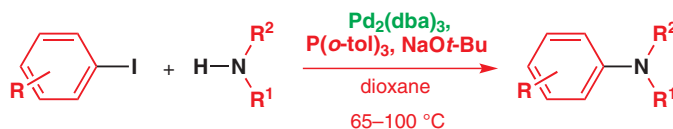
The mechanism is thought to be stepwise (in other words, not a real cycloaddition at all) with conjugate addition of the carbanion followed by attack of the resulting enolate on the π-allyl palladium unit to form a new five-membered ring having an *exo* methylene group.



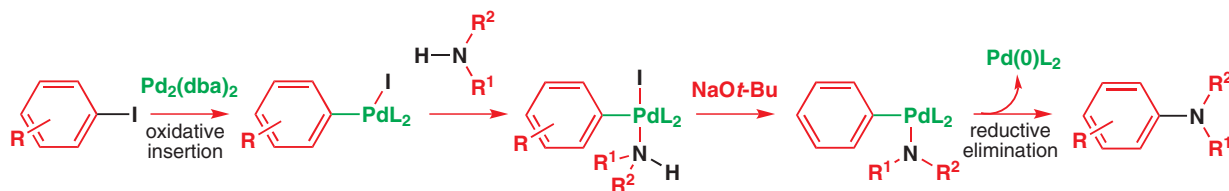
Interactive mechanism for the trimethylenemethane 'cycloaddition' catalytic cycle

Palladium-catalysed amination of aromatic rings

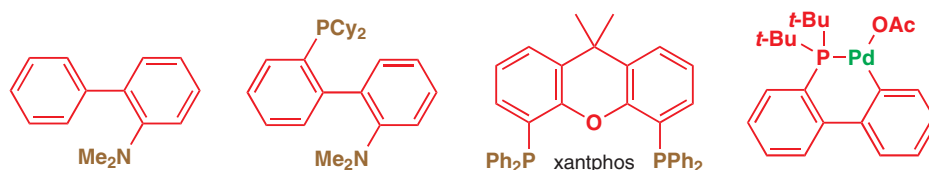
You've seen that palladium catalysis helps form carbon–carbon bonds that are difficult to make using conventional reactions. It can also help form carbon–heteroatom bonds that are difficult to make, and you have already seen some examples in the reactions of π-allyl complexes. Work starting in the 1990s by Buchwald and Hartwig has shown that Pd can be used to promote nucleophilic substitution at a vinylic or aromatic centre—a reaction which would not normally be possible. For example, aromatic amines can be prepared directly from the corresponding bromides, iodides, or triflates and the required amine in the presence of palladium(0) and a strong alkoxide base.



The mechanisms and catalysts used in this 'Buchwald–Hartwig' chemistry mirror those of coupling reactions involving oxidative addition, transmetalation, and reductive elimination. The first step, as usual, is oxidative insertion of Pd(0) into the aryl–halogen bond. The Pd(II) complex now adds the amine so that both coupling partners find themselves bonded to the same palladium atom. The base eliminates H–I from the complex and reductive elimination forms the Ar–N bond.

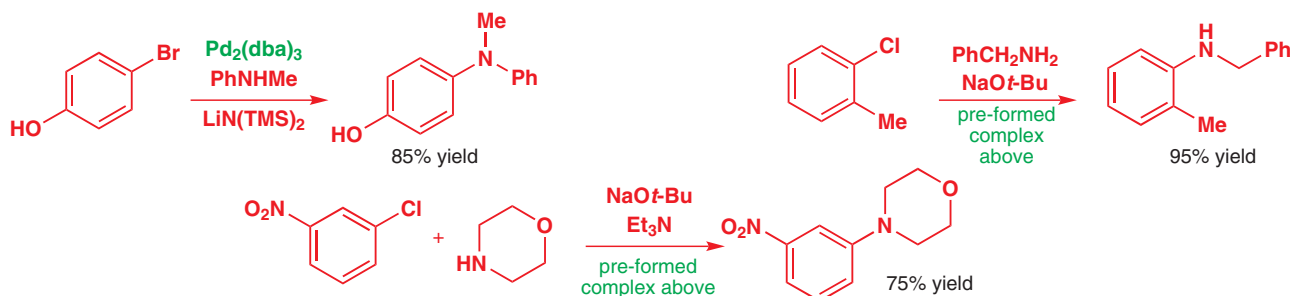


Various bases, such as *t*-BuONa, MeONa, LiN(TMS)₂, or K₂CO₃, have been successful and some of the most successful ligands (coordinating groups shown in brown) are shown below. The fourth structure is a preformed complex used in catalytic amounts.

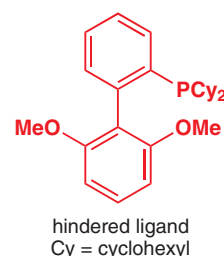
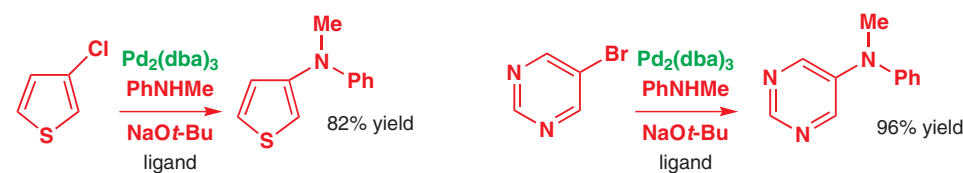


➔ If you are interested in reading more on the design and choice of these ligands, turn to the Further reading section at the end of the chapter.

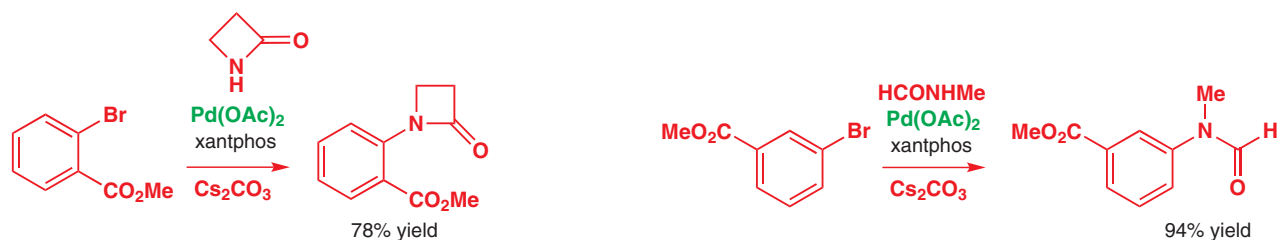
The range of compounds which can be made is very great: both electron-withdrawing and electron-donating substituents are acceptable; hindered compounds or those with acidic hydrogens such as phenols are tolerated. Even aryl chlorides, which are much cheaper than bromides or iodides, can also be successful.



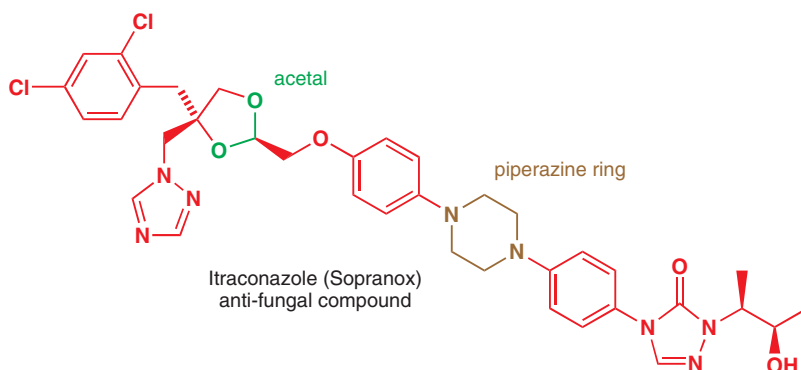
Aromatic heterocyclic halides also work well whether they are electron-deficient or electron-rich. These couplings use the more hindered ligand shown in the margin.



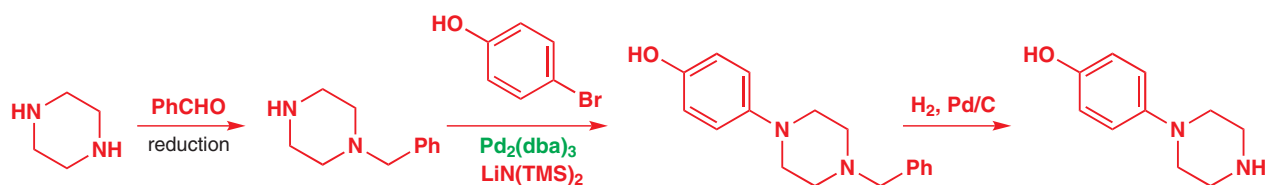
It is tempting to view the amine as the 'nucleophile' in these reactions but it is clear that nucleophilicity has little to do with it as amides also couple to aromatic rings under similar conditions. The ability to act as a ligand for palladium is the important thing. The ligand xantphos (see above) is used in these reactions and again the nature of the substituents on the benzene ring is of little account. Even strained azetidines react well.



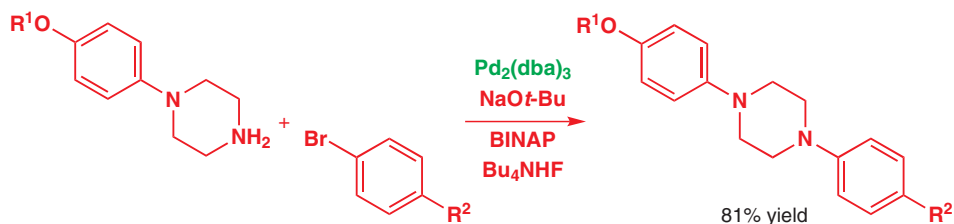
These reactions have been very widely used in the pharmaceutical industry in the making of medicinal compounds. When Sepracor wanted to make their anti-fungal compound itraconazole, it was obvious that they should make the two ends with stereochemistry and join them together with a central achiral section. Right in the middle is a piperazine ring joined to two different benzene rings, one connected through O and one through N. The C–N coupling chemistry of Buchwald and Hartwig could have been made for this problem.



We have already seen that *p*-bromophenol can be joined to an amine with palladium catalysis, so it should be easy to join it to piperazine. However, there is a potential problem of selectivity: we want to add this benzene ring just once, and the way to do this is to protect one nitrogen atom by reductive amination with benzaldehyde. The remaining NH group can then be coupled to the aromatic ring and the benzyl group removed by hydrogenation.



The workers at Sepracor then added the left-hand end of the molecule (we shall call this R^1) to the free OH group. The other aromatic ring, already functionalized with the right-hand end of the molecule (we shall call this R^2) was coupled as its bromide to the free NH group by a second Buchwald–Hartwig amination reaction process. It's easy to see how this chemistry simplifies the assembly of such a large and complex molecule.



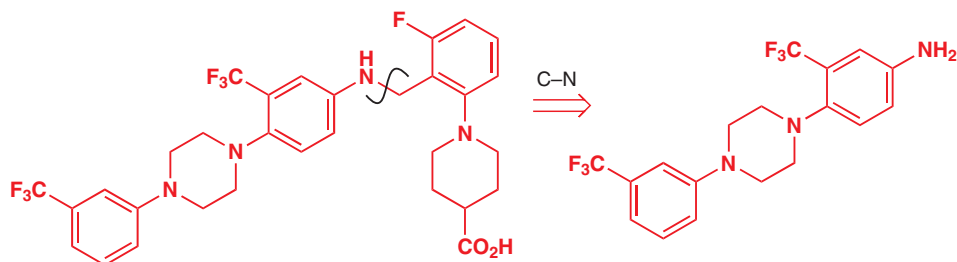
➡ The bisphosphine BINAP is shown on p. 319. It is a chiral compound, but that is irrelevant to its use here.

Nucleophilic aromatic substitution and palladium catalysis compared

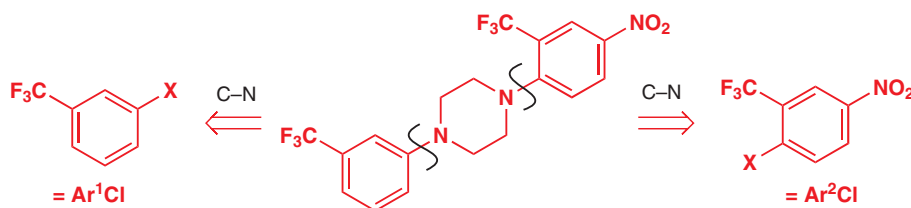
You will have noticed that Buchwald–Hartwig chemistry accomplishes the same as nucleophilic aromatic substitution (S_NAr , Chapter 22): the replacement of a halogen by a nucleophile. So what are the differences?

	S_NAr	Buchwald–Hartwig
the leaving group	$F > Cl > Br > I$ fluoride is not the best leaving group but it accelerates the addition	$I > Br > Cl \gg F$ iodide is best at the oxidative addition step but chloride will do and aryl chlorides are cheaper
regiochemistry	there must be an electron-withdrawing group <i>ortho</i> or <i>para</i> to the halide	any substitution pattern acceptable

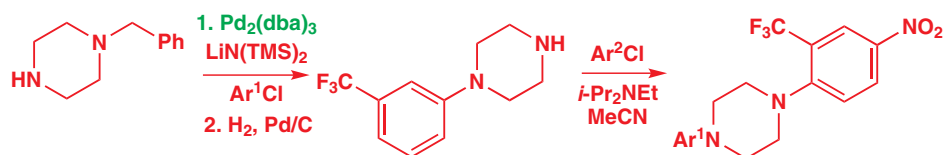
The synthesis of a drug to control blood clotting gives us the opportunity to review both methods. This compound also has a central piperazine ring and disconnection of the right-hand side chain reveals an amine that could be functionalized by alkylation with a suitable benzylic halide or reductive amination.



A standard way to make aromatic amines is by nitration and reduction (Chapter 21) so we can think of making this aminobenzene from the nitrobenzene below. Now we can disconnect the two C–N bonds with the idea of putting a halide (X) at the point of substitution in each aromatic coupling partner.



The substituents on the right-hand ring are both electron withdrawing and are *ortho* and *para* to the leaving group. As you know from Chapter 22, this is perfect for ordinary nucleophilic aromatic substitution—so much so that chloride is a good enough choice and it is not necessary to use fluoride. The left-hand ring has again a good electron-withdrawing substituent but it is *meta* to the halide and so nucleophilic aromatic substitution will not work. Palladium catalysis is needed. Chemists at Berlex Biosciences chose to introduce the left-hand ring first.



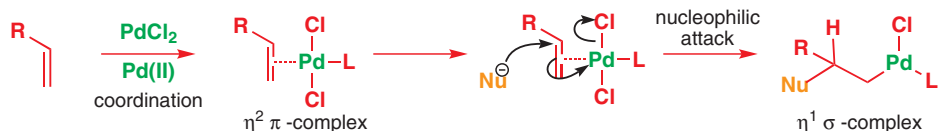
Alkenes coordinated to palladium(II) are attacked by nucleophiles

Now for another case where a transition metal catalysis facilitates a reaction that would not occur under normal conditions: nucleophilic attack on an isolated double bond. Usually alkenes react with nucleophiles only when conjugated with an electron-withdrawing group. But coordination of an electron-rich alkene to a transition metal ion such as palladium(II) changes its reactivity dramatically: electron density is drawn towards the metal and away from the π orbitals of the alkene. This leads to activation towards attack by nucleophiles, just as in conjugate addition, and unusual chemistry follows. Unusual, that is, for the alkene; the palladium centre behaves exactly as expected.

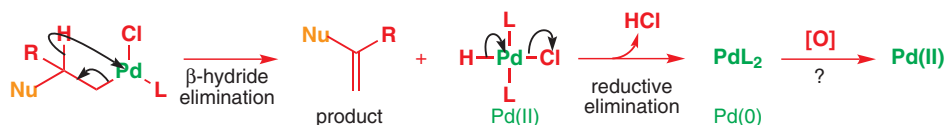


■ This regioselectivity is not the same as in the Heck reaction, where attack mostly occurs at the end of the alkene. Internal nucleophiles transferred from the palladium to the alkene usually prefer the terminal position of the alkene but external nucleophiles usually prefer the more substituted end.

Many nucleophiles, such as water, alcohols, and carboxylates, are compatible with an alkene–Pd(II) complex and can attack the complexed alkene from the side opposite the palladium. The attack of the nucleophile is regioselective for the more substituted position. This parallels attack on bromonium ions but is probably governed by the need for the bulky palladium to be in the less hindered position.



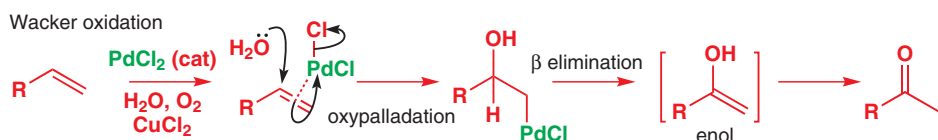
The resulting Pd(II) σ -alkyl species decomposes by β -hydride elimination to reveal the substituted alkene. Reductive elimination of a proton and the leaving group, usually chloride, leads to palladium(0). The weakness of this reaction is that the catalytic cycle is not complete: Pd(II), not Pd(0), is needed to complex the next alkene.



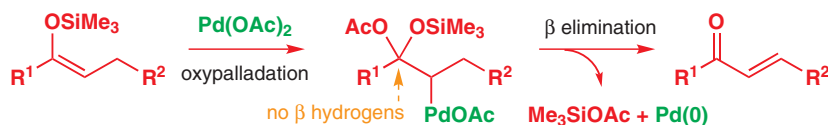
There are two solutions to this problem. We could use stoichiometric Pd(II) but this is acceptable only if the product is very valuable or the reaction is performed on a small scale. It is better to use an external oxidant to return the palladium to the Pd(II) oxidation state so that the cycle can continue. Air alone does not react fast enough (even though Pd(0) must be protected from air to avoid oxidation) but, in combination with copper(II) chloride, oxygen completes the catalytic cycle. CuCl_2 oxidizes Pd(0) to Pd(II) and is itself oxidized back to Cu(II) by oxygen, ready to oxidize more palladium.

Oxypalladation and the Wacker oxidation

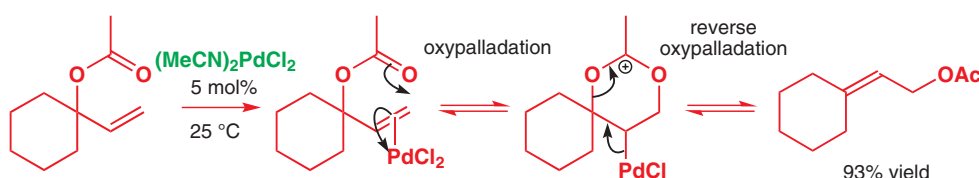
This combination of reagents has been used to oxidize terminal vinyl groups to methyl ketones and is known as the **Wacker oxidation**. The nucleophile is simply water, which attacks the activated alkene at the more substituted end in an *oxypalladation* step. β -Hydride elimination from the resulting σ -alkyl palladium complex releases the enol, which is rapidly converted into the more stable keto form. Overall, the reaction is a hydration of a terminal alkene that can tolerate a range of functional groups.



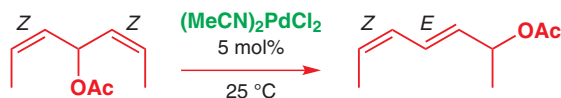
A related reaction is the oxidation of silyl enol ethers to enones. This requires stoichiometric palladium(II), although reoxidation of Pd(0) with benzoquinone can cut that down to about half an equivalent. The reaction provides a valuable way of turning regioselective methods for making silyl enol ethers (Chapter 20) into regioselective methods for oxidizing ketones to enones. The first step is again oxypalladation and β elimination puts the alkene in conjugation with the ketone: there are no β hydrogens on the other side.



An example of catalytic oxypalladation is the rearrangement of allylic acetates with Pd(II). The reaction starts with oxypalladation of the alkene and it is the acetate already present in the molecule that provides the nucleophile to attack the alkene. The intermediate can reverse the oxypalladation in either direction and the product is whichever allylic acetate has the more substituted alkene. In this case, trisubstituted beats monosubstituted easily.

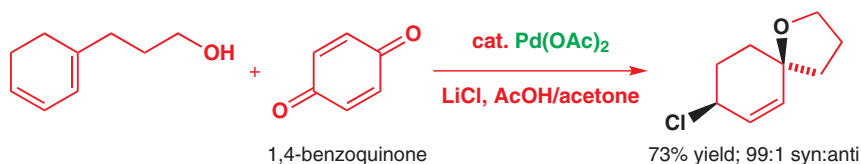


The reaction is *E*-selective, which means that a simple synthesis of an *E,Z*-diene is possible from the symmetrical acetate with two *Z*-allylic alkenes. The one that rearranges goes *E* and the one that stays behind remains *Z*. The driving force for this rearrangement, from one disubstituted alkene to another, is establishment of conjugation.

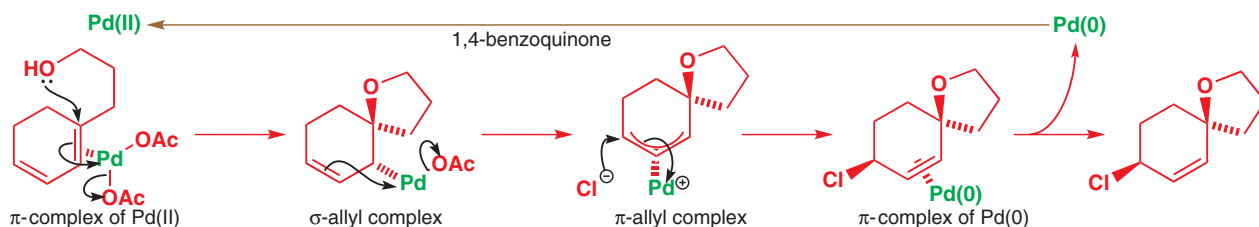


Alcohols and amines as intramolecular nucleophiles

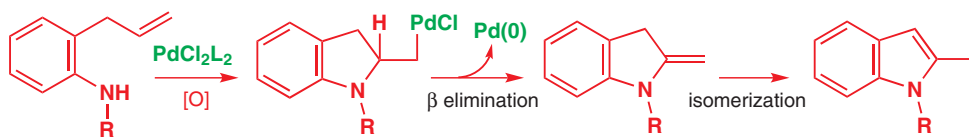
Cyclic ethers and amines can be formed with an intramolecular alcohol or amine nucleophile. Stoichiometric palladium can be avoided by using benzoquinone as the stoichiometric oxidant with a catalytic amount of palladium. In this example intramolecular oxypalladation of a diene is followed by attack of an external nucleophile on a π -allyl complex.



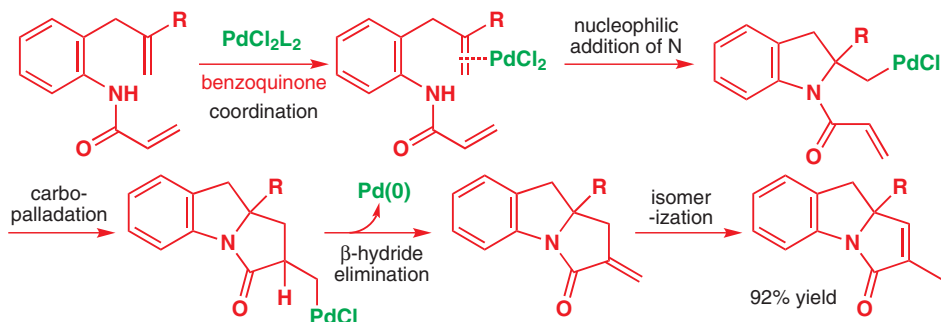
Palladium coordinates to one face of the diene, promoting intramolecular attack by the alcohol on the opposite face. The resulting π -allyl palladium can form a π -allyl complex with the palladium on the lower face simply by sliding along to interact with the double bond. Nucleophilic attack of chloride from the lithium salt then proceeds in the usual way on the face opposite palladium. The overall addition to the diene is therefore *cis*.



Nitrogen nucleophiles also attack alkenes activated by Pd(II), and benzoquinone can again act as a reoxidant, allowing the use of catalytic quantities of palladium. The mechanism follows the same pattern as for oxygen nucleophiles, and a final isomerization produces the most stable regioisomer of product. In this example the product is an aromatic indole, so the double bond migrates into the five-membered ring.



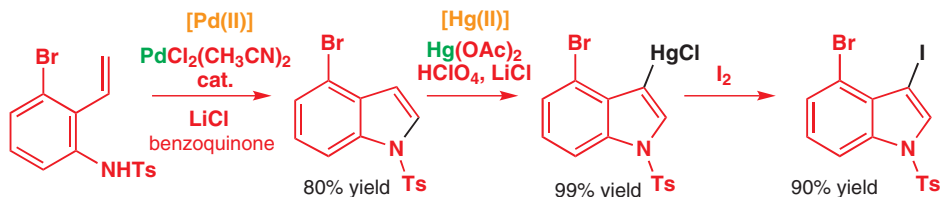
If the substrate lacks a hydrogen suitable for β elimination and there is another alkene present in the molecule, the σ -alkyl palladium intermediate can follow the Heck pathway to form a bicyclic structure in a tandem reaction sequence. Once again, the final step is a palladium-hydride-mediated isomerization to give the endocyclic alkene.



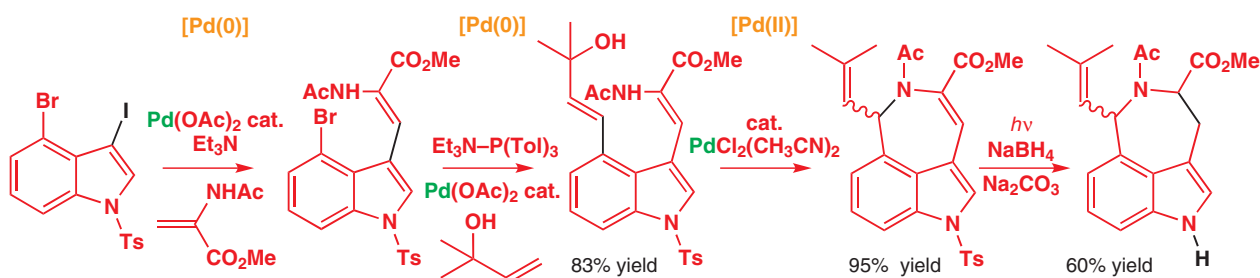
Palladium catalysis in the total synthesis of a natural alkaloid

We take our leave of palladium by presenting a synthesis of an alkaloid, *N*-acetyl clavicipitic acid methyl ester, by Hegedus. The power of organometallic chemistry is illustrated in five of the steps in this seven-step process (the metals are highlighted in orange). Each of the organometallic steps catalysed by Pd(0) or Pd(II) has been described in this chapter. The overall yield is 18%, a remarkably good result for a molecule of such complexity.

The first step is to make an indole by Pd(II)-catalysed cyclization in the presence of benzoquinone as reoxidant. The nucleophilic nature of the 3-position of the indole (Chapter 30) was exploited to introduce the required iodide functionality. Rather than direct iodination, a high-yielding two-step procedure involving mercuriation followed by iodination was employed.



Aryl iodides are more reactive towards oxidative addition than aryl bromides, and a selective Heck coupling (without phosphine ligands) with an unsaturated side chain left the bromide in place. A second Heck reaction of this bromide with an allylic alcohol was used to introduce a second side chain. Cyclization of the amide on to the allylic alcohol was achieved with palladium catalysis, not as might have been expected with palladium(0) but instead with palladium(II), to produce the seven-membered ring. Finally, the conjugated double bond was reduced and the sulfonamide removed under photolytic conditions.

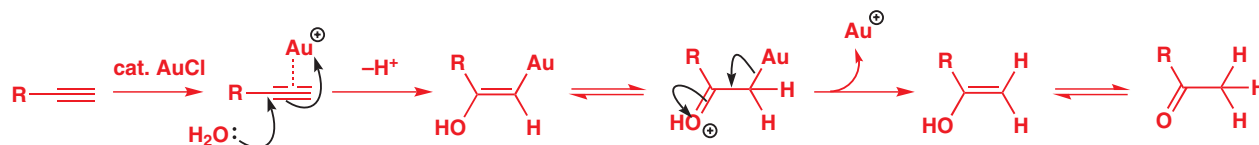


An overview of some other transition metals

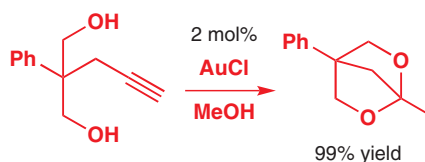
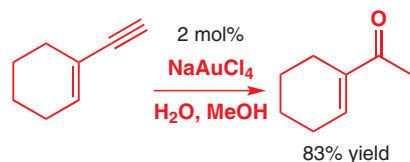
Some metals—palladium chief among them—see continual service in catalysis but others have their day and then fall out of favour when better alternatives become available. Tin is less popular now than it was 20 years ago because of its toxicity. A more serious case is mercury. Mercury(II) is an excellent catalyst for the addition of water to alkynes. But mercury is very toxic indeed, and the last ten years have seen its role largely superseded by gold. Do not recoil at the expense! Gold is expensive on the scale used to make rings, plates, medals, and coins, but here it is used in only catalytic quantities. Gold is in fact less expensive than palladium, rhodium, or ruthenium. Part of the age-old appeal of gold is its unreactivity as a metal: it is very stable but it does form Au(I) and Au(III) salts such as AuCl and AuCl₃. Both are available commercially and are generally used as their phosphine complexes.

Gold: activating alkynes

Au(I) and Au(III) form cationic π complexes with alkynes and these react with nucleophiles of many kinds. With water the result is simple: it adds to the more substituted end of the alkyne and the net result is hydration to give a ketone.



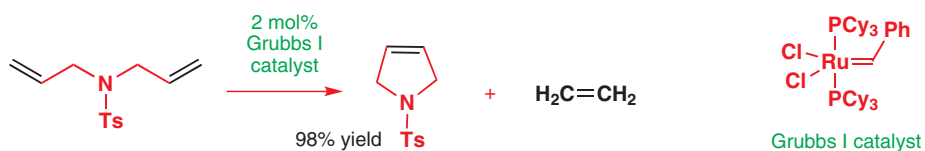
This simple reactivity can be developed in many more elaborate ways you can read about elsewhere, but simple examples include hydration of an enyne to form a conjugated ketone and the capture of the ketone by intramolecular acetal formation. The details give you an idea of reagents, solvents, and yields.



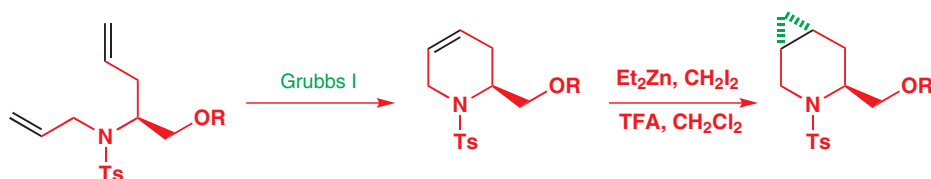
Ruthenium: alkene (olefin) metathesis

The theme of this chapter is that transition metals let you do things to organic molecules which are unthinkable without them. Nowhere is this more true than in metathesis reactions, and we finish the chapter with a reminder of the power of the ruthenium catalysts we introduced in Chapter 38. There we discussed the carbene-based mechanism of the reaction, and we showed you some simple examples such as this cyclization of a symmetrical amine to give a five-membered heterocycle using a catalytic amount of the ruthenium complex known as Grubbs I catalyst.

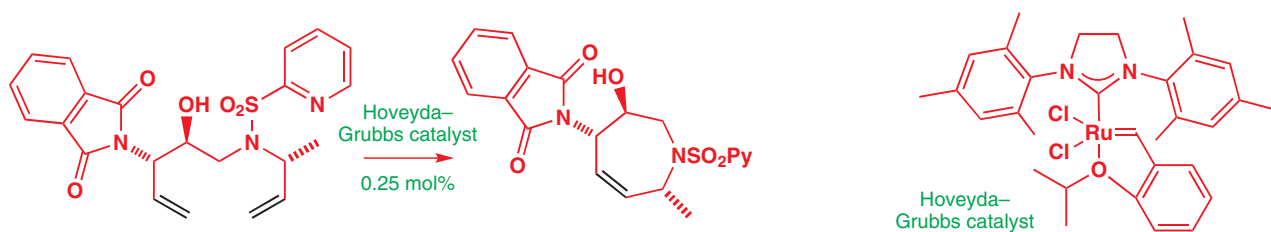
➡ The structures of three important ruthenium complexes used as catalysts for metathesis are given on p. 1025.



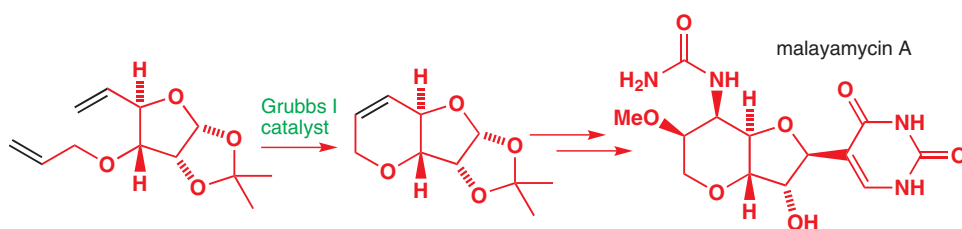
The synthesis of a sleep-inducing drug by GlaxoSmithKline in their laboratories at Verona used a very similar metathesis, although on an unsymmetrical amine and giving a six-membered heterocycle. The starting material is also a single enantiomer and the stereochemistry is important as the cyclopropane, introduced by a Simmons–Smith reaction (Chapter 38), must be be on the opposite face of the six-membered ring to the side chain.



At another GlaxoSmithKline site, in the USA, the development of a drug for osteoporosis and osteoarthritis required a seven-membered heterocycle with two controlled chiral centres. This time the Hoveyda–Grubbs catalyst had to be used but the loading is very low indeed. Notice also that a free OH group does not interfere.



Our third example comes from Syngenta's crop protection laboratory in Basel. It is another cyclization but this time to form an oxygen heterocycle with four chiral centres. The final product of this synthesis is malayamycin A, a natural fungicide found in bacteria. The metathesis step is early in the synthesis and you will notice that the alkene formed in this cyclization is used to provide two more chiral centres in malayamycin.



In the next chapter you will see more ways in which ruthenium—along with osmium, titanium, rhodium, and others—can be used to solve the challenges of synthesis as we look at ways of making molecules as single enantiomers.

Further reading

Most textbooks of organometallic chemistry favour the inorganic approach of facts rather than explanation. There are usually plenty of structures and catalytic cycles but very few mechanisms. However, two brief introductions that might help you are: M. Bockmann, *Organometallics 1 and 2*: Oxford Primers, OUP, Oxford, 1994. A book that does contain mechanisms of a number of the reactions in this book, as well as others, is P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007. Probably the best comprehensive account is J. Hartwig, *Organotransition Metal Chemistry*, University Science Books, New York, 2010.

The references to the examples of drug synthesis by metathesis are: W. M. Maton and GlaxoSmithKline group in Verona, *Organic Process Research and Development*, 2010, **14**, 1239; H. Wang and GlaxoSmithKline group in King of Prussia, Pennsylvania, *Organic*

Process Research and Development, 2008, **12**, 226; O. Loiseleur and Syngenta group at Basel, *Organic Process Research and Development*, 2006, **10**, 518.

Organic Syntheses are a good source of ways to make reagents and ways to carry out reactions. Comins' reagent is featured in *Organic Syntheses*, 1997, **74**, 77.

Leading references for the Buchwald and Hartwig chemistry: J. F. Hartwig and group, *Angew. Chem. Int. Ed.*, 2005, **44**, 1371; S. L. Buchwald and group, *Organic Letters*, 2005, **7**, 3965. Gold chemistry is reviewed by A. Fürstner and P. W. Davies, *Angew. Chem. Int. Ed.*, 2007, **46**, 3410. The drug syntheses are from C. H. Senanayake and group, *Tetrahedron: Asymmetry*, 2003, **14**, 3487; B. Ye and group, *Bioorg. and Med. Chem. Lett.*, 2004, **14**, 761.

The new gold chemistry of alkynes and alkenes is described in a long review H. C. Shen, *Tetrahedron*, 2008, **64**, 3885.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>